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1

Abstract

Selective-area growth (SAG) in metal–organic vapor-phase epitaxy (MOCVD) has been used to grow layers of semiconductor material on planer substrate patterned with silicon masks. In the vicinity of the mask, growth is modified because precursors that are not incorporated on masked regions are redistributed toward exposed crystal surfaces. Classical vapor-phase diffusion (VPD) models were demonstrated using a diffusion equation in an stagnant layer to predict growth thickness by parameter fitting, but they do not resolve reactor-scale advection or coupled multicomponent transport.

We implemented a multi-species Lattice-Boltzmann Method (LBM) on a D2Q9 lattice with a semi-implicit BGK collision scheme in transformed variables, combined with a coupling-aware equilibrium formulation and a local post-streaming Maxwell–Stefan flux correction. Reactive Robin boundary conditions are imposed on unmasked wafer regions, while masked regions are treated as inert no-flux boundaries. The relevant transport regimes are reproduced by matching the Reynolds and P'eclet numbers at two simulation scales. At reactor scale, the model captures advection-dominated concentration boundary-layer development above the wafer, including precursor depletion over the wafer and the formation of a downstream product wake. At micro-scale, transport is diffusion-dominated, and the predicted near-surface reactant-supply patterns qualitatively follow classical Laplace/VPD solutions while reproducing strong concentration variations associated with mask-pattern interference. Comparison with experimental InGaAsP characterization shows partial agreement in indium-composition trends, but also highlights clear discrepancies.

A central result is that the uptake parameter estimated from the measured InP growth rate differs by about three orders of magnitude from the effective diffusion lengths obtained by our LBM model or VPD parameter fit from literature. This suggests that a single-step surface-reaction description is insufficient, and that crystal growth may instead be limited by low-concentration intermediate species generated through temperature-dependent gas-phase and surface chemistry. Overall, this work shows that the multi-species LBM provides a good understanding of advection-diffusion transport at reactor scale, and is able to reproduce VPD model's prediction at mask scale ($10^{-5}\mathrm{m}$); however, a true understanding to the system will require thermal coupling, variable properties, and temperature-dependent reaction chemistry.

Nomenclature

Symbol	Meaning
Indices, sets, lattice directions	
$s \in \{1, \dots, N\}$	Species index.
c	Species index used in sums (cross-coupling with species s).
N	Number of species.
$i \in \{0, \dots, 8\}$	D2Q9 direction index.
$\bar{i}$	Opposite direction of i.
$\alpha \in \{x, y\}$	Cartesian component index.
$\mathcal{I}_{\text{in}} = \{2, 3, 4\}$	Inlet incoming direction set (as used in your BC derivations).
$\mathcal{I}_{\text{bot}} = \{1, 2, 8\}$	Bottom-wall incoming direction set.
$\mathcal{I}_{\text{top}} = \{4, 5, 6\}$	Top-wall incoming direction set.
$\mathcal{I}_{\text{out}} = \{6, 7, 8\}$	Outlet incoming direction set.
Space, time, geometry	
$\mathbf{x},\ (x, y)$	Spatial coordinate (vector / components).
t	Time.
Δx	Lattice spacing (also used in half-cell wall reconstructions).
Δt	Time step.
$n_x,\ n_y$	Grid sizes in x and y.
L	Characteristic length scale (dimensionless numbers).
U	Characteristic velocity scale (dimensionless numbers).
$w,\ h$	Chamber cross-section dimensions; $A = wh$.
A	Cross-sectional area.
V	Volume (residence time estimate).
D_h	Hydraulic diameter.
L_w	Wafer-scale characteristic length.
L_μ	Near-surface micro-layer length scale.
H	Channel height in Couette–Poiseuille profile.
$\xi,\ L_\xi$	Coordinate and associated domain length for shear-wave test.
LBM distributions, moments, velocities	
$f_{s,i}(\mathbf{x}, t)$	Species-s distribution in direction i.
$g_{s,i}$	Transformed distribution (semi-implicit / trapezoidal BGK).
$g^{+}_{s,i}$	Post-collision transformed population.
$f^{eq}_{s,i}$	Equilibrium distribution.
ρ_s	Partial density / concentration of species s.
$\rho_{\text{mix}} = \sum_{s=1}^{N} \rho_s$	Mixture density.
$\mathbf{u}_s$	Species velocity.
$\mathbf{u}$	Mixture mass-averaged velocity.
$\mathbf{v}$	Mixture mole-averaged velocity.
$\mathbf{q}_s \equiv \rho_s \mathbf{u}_s$	Species momentum / advective mass flux (your definition).
$\tilde{\mathbf{q}}_s$	Provisional momentum from streamed transformed populations.
$\mathbf{j}_s = \sum_i \mathbf{c}_i f_{s,i}$	Species flux moment used in inlet discussion.
$\mathbf{u}^*_s$	Coupling-modified velocity (via χ_{sc}).
$\mathbf{u}_w$	Wall velocity (e.g., moving top wall; often $\mathbf{u}_w = \mathbf{0}$ at bottom).
D2Q9 lattice parameters	
$\mathbf{c}_i$	Discrete velocity vector for direction i.
$c_{x,i},\ c_{y,i}$	Components of $\mathbf{c}_i$.
w_i	D2Q9 lattice weights.
c_s^2	Lattice speed of sound squared (D2Q9: $c_s^2 = 1/3$).

Symbol	Meaning
Composition variables	
$Y_s = \rho_s/\rho_{\mathrm{mix}}$	Mass fraction.
$y_s = n_s/n$	Mole fraction.
$n_s,\ n$	Molar density of species s; total molar density $n = \sum_s n_s$.
M_s	Molar mass of species s.
$M_{\mathrm{mix}} = \left(\sum_s Y_s/M_s\right)^{-1}$	Mixture molar mass.
Maxwell–Stefan coupling & semi-implicit collision	
B_{sc}	Maxwell–Stefan friction/coupling coefficient.
B_{ss}	Diagonal/self friction coefficient.
β	Multiplier scaling the B_{sc} matrix (tunes lattice diffusivities).
χ_{sc}	Dimensionless coupling coefficient used in $\mathbf{u}_s^*$.
$\lambda_s = c_{\mathrm{mix}}^2 B_{ss}$	Relaxation strength in trapezoidal formulation.
p_{mix}	Mixture pressure surrogate.
$c_{\mathrm{mix}}^2 = p_{\mathrm{mix}}/\rho_{\mathrm{mix}}$	Effective mixture sound-speed-like quantity.
ϕ_s	Species parameter in equilibrium (e.g., $\phi_s = \phi_1 M_1/M_s$).
θ	Trapezoidal parameter (typically $\theta = \frac{1}{2}$).
$\omega,\ \omega_s^{\mathrm{eff}} = \lambda_s/(1+\theta\lambda_s)$	BGK relaxation rate; effective rate in semi-implicit BGK.
Transport coefficients	
D_s	Molecular diffusion coefficient (physical).
D_s^{LBM}	Lattice diffusion coefficient.
$D_{s,\mathrm{req}}^{\mathrm{LBM}}$	Required lattice diffusivity to match target $\mathrm{Pe}_s^{\mathrm{phys}}$.
μ_{dyn}	Dynamic viscosity (physical).
$\nu = \mu_{\mathrm{dyn}}/\rho$	Kinematic viscosity (physical).
$\nu_{\mathrm{LBM}} = c_s^2\left(\frac{1}{\omega^{\mathrm{eff}}} - \frac{1}{2}\right)$	Lattice kinematic viscosity relation.
$\nu_s = \frac{1}{3}\left(\frac{1}{\omega_s^{\mathrm{eff}}} - \frac{1}{2}\right)$	Species-wise lattice viscosity (as written in methods).
Dimensionless groups & mass transfer	
$\mathrm{Re} = UL/\nu$	Reynolds number.
$\mathrm{Sc}_s = \nu/D_s$	Schmidt number for species s.
$\mathrm{Pe}_s = UL/D_s = \mathrm{Re}\,\mathrm{Sc}_s$	Péclet number for species s.
$\mathrm{Sh}_s = k_{m,s}L/D_s$	Sherwood number for species s.
$k_{m,s}$	External mass-transfer coefficient.
$\mathrm{Re}_x,\ \delta_{99}(x)$	Local Reynolds number; boundary-layer thickness estimate (99% criterion).
$\mathrm{Da} = k_{\mathrm{rxn}}/k_{m,s}$	Damköhler number (as used in your estimate).
k_{rxn}	Effective reaction rate coefficient.
Flow/throughput and gas properties	
$\dot{V},\ \dot{V}_{\mathrm{std}}$	Volumetric flow rate (actual; standard conditions).
$T,\ T_{\mathrm{std}};\ p,\ p_{\mathrm{std}}$	Temperature / pressure (actual; standard).
R	Gas constant.
ρ	Gas density (physical).
$U_{\mathrm{ch}},\ U_w,\ U_\mu$	Velocity estimates (chamber-, wafer-, micro-layer scales).
$\dot{\gamma}$	Shear rate used in micro-layer scaling.
$\tau_{\mathrm{ch}} = V/\dot{V}$	Residence time estimate.
Boundary-condition variables	
$\rho_{s,\mathrm{in}}$	Prescribed inlet density for species s (enforced-equilibrium inlet).
$C_s^{\mathrm{in}}(y)$	Prescribed inlet concentration profile (species-wise).
$\mathbf{u}^{\mathrm{in}}(y),\ u_x(y), u_y(y)$	Prescribed inlet velocity profile / components.
ρ_{tot}	Total inlet density (Zou/He-style treatment).
∂_n	Inward normal derivative at wall (into the domain).

Symbol	Meaning
$C_s,\ C_{f,s},\ C_{w,s}$	Concentration field; first-fluid-node value; effective wall value (anti-bounce-back).
b_1, b_2, b_3	Robin coefficients in $b_1\,\partial_n C_s + b_2\,C_s = b_3$.
J_s	Wall flux of species s.
Reaction / stoichiometry at bottom wall	
$J_{\mathrm{TMIn}},\ J_{\mathrm{PH_3}},\ J_{\mathrm{CH_4}}$	Wall fluxes (TMIn consumed; PH_3 consumed; CH_4 produced).
$\dot{h}$	InP growth rate (thickness/time).
$\rho_{\mathrm{InP}},\ M_{\mathrm{InP}}$	InP solid density; InP molar mass.
c_{TMIn}	TMIn concentration used in $k_{\mathrm{rxn}} \approx J_{\mathrm{InP}}/c_{\mathrm{TMIn}}$.
Validation-case variables	
$\sigma(t),\ \sigma_0$	Gaussian standard deviation (tracer diffusion); initial value.
D	Diffusivity from $\sigma^2(t) = \sigma_0^2 + 2Dt$.
$u_0,\ U_0$	Initial amplitude (shear-wave; Taylor–Green vortex).
$k = 2\pi/L_\xi$	Wavenumber used in decay tests.

2

Introduction

2.1. Background: selective-area growth in MOCVD

Metal–organic chemical vapor deposition (MOCVD), also referred to as metal–organic vapor-phase epitaxy (MOVPE), is a key fabrication method for III–V heterostructures used in photonics and high-speed electronics. In a typical MOVPE process, organometallic precursors supply group-III elements (e.g., TMIn, TMGa, TMAl), while hydride sources supply group-V elements (e.g., AsH_3, PH_3), transported by a carrier gas (often H_2). Because the process occurs in a flowing reactor and involves coupled transport and surface reactions, predictive modeling is central to process design and to reproducible device performance.

A particularly important variant is *selective-area growth* (SAG), where a dielectric mask (commonly SiO_2) is patterned on the substrate to inhibit epitaxy on masked regions while allowing growth in openings. SAG enables in-plane integration by locally modifying growth rate, thickness, and composition using only the mask geometry. Experiments have consistently shown that, compared to planar growth, selectively grown layers become *thicker* and may become *compositionally shifted* (e.g., indium enrichment) near mask edges or within narrow openings because reactants that would otherwise be consumed on the masked area are redistributed to the exposed crystal regions [1]. For practical SAG, high *selectivity* is crucial: any parasitic deposition on the dielectric mask not only wastes reactants but also makes the effective supply to openings unpredictable [2].

2.2. Diffusion-equation view of SAG: vapor-phase diffusion models

2.2.1. Physical picture: stagnant layer and lateral precursor redistribution

A widely used starting point for SAG modeling is the *vapor-phase diffusion* (VPD) model. The central assumptions are: (i) laminar reactor flow, (ii) a stagnant boundary layer adjacent to the wafer in which transport is diffusion-dominated, (iii) no incorporation on the mask (perfect selectivity), and (iv) incorporation on exposed crystal surfaces through a first-order surface reaction [1, 3]. Within this framework, lateral redistribution of group-III precursors occurs primarily in the gas phase above the wafer, and long-range thickness and composition perturbations can be described without explicitly resolving the full reactor-scale flow.

2.2.2. From diffusion equation to Laplace equation

Let $c(\mathbf{x})$ denote the gas-phase concentration of a reactive species within the stagnant layer. Neglecting homogeneous reactions inside this layer and assuming quasi-steady transport, species conservation gives

$$\nabla \cdot (D \nabla c) = 0, \tag{2.1}$$

and for constant diffusivity D this reduces to Laplace's equation,

$$\nabla^2 c = 0. \tag{2.2}$$

This is the core PDE solved in classical SAG VPD models [1–3].

2.2.3. Boundary conditions: mask (no-flux) and crystal (reactive sink)

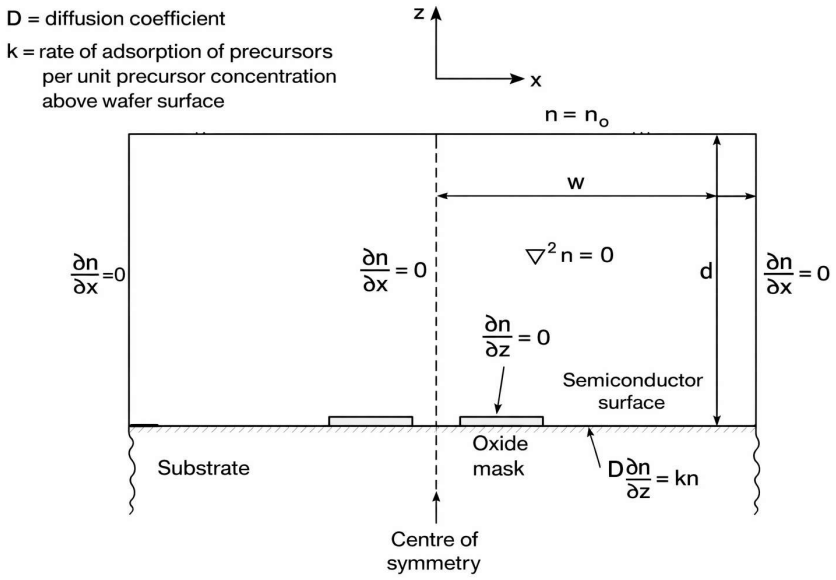


Figure 2.1: Schematic of the diffusion–adsorption model for precursor transport above a patterned semiconductor surface with an oxide mask. The widths of the masks and the gap between them are on the order of $\sim 30\ \mu$m.

To close Eq. (2.2), VPD models impose boundary conditions that encode selectivity and surface kinetics. On the dielectric mask, where incorporation is suppressed, the normal diffusive flux is set to zero:

$$\left.\frac{\partial c}{\partial n}\right|_{\text{mask}} = 0. \tag{2.3}$$

On the exposed semiconductor, incorporation is commonly modeled as a first-order sink proportional to the local near-surface concentration (a Langmuir-isotherm-like condition in the low-coverage limit):

$$-D\left.\frac{\partial c}{\partial n}\right|_{\text{crystal}} = k\,c \qquad \Leftrightarrow \qquad \left.\frac{\partial c}{\partial n}\right|_{\text{crystal}} = -\frac{k}{D}\,c, \tag{2.4}$$

where k is an effective surface reaction (sticking/incorporation) rate constant with the dimension of m/s [1, 3].

At the top of the stagnant layer (height H above the wafer), a fixed concentration is typically prescribed, representing the well-mixed bulk flow above the stagnant layer:

$$c(z = H) = c_0. \tag{2.5}$$

Depending on the geometric idealization, lateral boundary conditions are taken as symmetry or periodicity conditions (e.g., zero normal derivative at the lateral boundaries of a repeated unit cell) [1, 3].

2.2.4. A key parameter: the effective diffusion length D/k

The ratio D/k has dimensions of length and plays the role of an *effective diffusion length* in VPD models: smaller D/k means stronger or more rapid surface consumption (sharper spatial decay), while larger

D/k spreads excess supply over longer distances [1]. By **fitting measured thickness** and composition profiles around masks, one can extract species-specific D/k values and then use them to predict growth enhancement and composition variations for new mask layouts [1, 2].

A common VPD output is the *growth-rate enhancement* (or normalized concentration) field,

$$R(\mathbf{x}) = \frac{c_{\text{eff}}(\mathbf{x})}{c_0}, \tag{2.6}$$

where c_{eff} is the near-surface concentration on the crystal relative to a planar reference [3]. Spatial variations of R directly translate into thickness nonuniformity, while different R_s among multiple group-III species translate into composition shifts in ternary and quaternary alloys.

2.3. Motivation for transport-resolved simulation in reactor-scale geometries

Diffusion-equation (VPD) models provide a compact and physically transparent description of how mask-induced concentration gradients translate into SAG thickness and composition variations. However, they rely on several simplifications, including stagnant flow, simplified single-species diffusion, and limited coupling between diffusion and advection. Fluid-dynamic methods, on the other hand, are capable of simulating the advection–diffusion transport of the system. Flow-coupled diffusion may add effects beyond the pure-diffusion result. Moreover, simulating a multi-species system enables the modeling of advection driven by diffusion, as well as chemical reactions in the gas flow.

2.4. Project scope

The main goal of this project is to investigate whether a multicomponent lattice Boltzmann approach can reproduce and extend the insights of classical VPD models for SAG. The focus is on gas-phase advection–diffusion of dilute precursors in an isothermal carrier flow, and on how masking perturbs the local near-surface reactant supply.

The work in this report develops an isothermal Maxwell–Stefan-coupled multi-species lattice Boltzmann solver on a D2Q9 lattice. A single-relaxation-time (BGK) collision model is used, together with a semi-implicit (trapezoidal) discretization for improved stability. The implementation is designed with GPU acceleration in mind and is documented as an accessible simulation toolbox.

On the modeling side, reactor-relevant boundary conditions are implemented, including laminar inflow profiles, no-slip walls, a simple outflow, and a segmented bottom boundary that alternates between reactive (unmasked) and inert (masked) regions. Surface incorporation is represented using linear Robin boundary conditions (first-order uptake), with a simplified stoichiometric coupling between reactants ($TMIn/PH_3$) and a product species (CH_4).

Simulations are performed at two scales: a reactor-scale 2D cross-section that captures advection-dominated transport above a wafer, and a near-surface micro-layer simulation resolving diffusion-dominated transport over a patterned mask. The implementation is validated with canonical diffusion and viscosity benchmarks, and the micro-layer predictions are compared against a diffusion-only (Laplace) reference model.

The main limitation is that the current implementation only describes an isothermal, constant-property system. Temperature gradients, variable material properties (e.g. T-dependent viscosity and diffusivity), variable-density effects, and buoyancy-driven convection are not modeled. Detailed gas-phase decomposition and multi-step surface chemistry are also left out of scope: reactions are represented only through boundary Robin conditions, and no homogeneous gas-phase reactions are solved.

3

Methods

3.1. Approach overview

We simulate multicomponent advection–diffusion in an isothermal gas mixture using a Maxwell–Stefan coupled lattice Boltzmann method (MS–LBM) [4–7]. The solver evolves one D2Q9 distribution function per species, $f_{s,i}(\mathbf{x}, t)$ for $s \in \{1, \dots, N\}$ and lattice directions $i \in \{0, \dots, 8\}$.

Maxwell–Stefan cross-diffusion enters the discrete scheme in two coupled steps:

1. a *coupling-aware equilibrium* built from a modified velocity $\mathbf{u}_s^*$ that embeds interspecies friction,
2. a *post-streaming flux correction* obtained by solving, at every lattice node, a local $N \times N$ linear system to enforce Maxwell–Stefan-consistent coupled momenta.

To improve stability when relaxation strengths vary with composition, collisions are implemented with a trapezoidal (semi-implicit) BGK discretization via a transformed distribution $g_{s,i}$.

3.2. Multi-component system and Maxwell–Stefan diffusion

3.2.1. Concentration measures, mixture velocities, and diffusion fluxes

We distinguish between mass- and molar-based composition measures. Given partial densities ρ_s and molecular weights M_s, define

$$Y_s \equiv \frac{\rho_s}{\rho_{\mathrm{mix}}}, \qquad \rho_{\mathrm{mix}} \equiv \sum_{s=1}^{N} \rho_s, \qquad \sum_{s=1}^{N} Y_s = 1, \tag{3.1}$$

and the molar densities $n_s = \rho_s / M_s$ with total molar density $n = \sum_s n_s$ and molar concentrations

$$y_s \equiv \frac{n_s}{n}. \tag{3.2}$$

Two mixture velocities are commonly used: the barycentric (mass-averaged) velocity and the molar-averaged velocity,

$$\mathbf{u} = \sum_{s=1}^{N} Y_s \mathbf{u}_s, \qquad \mathbf{v} = \sum_{s=1}^{N} y_s \mathbf{u}_s. \tag{3.3}$$

Correspondingly, one may define mass- and molar-diffusion fluxes relative to these mixture velocities.

$$y_s = \frac{Y_s / M_s}{\sum_{c=1}^{N} Y_c / M_c}. \tag{3.4}$$

3.2.2. Maxwell–Stefan

For binary mixtures under restrictive assumptions, a Fick law $J = D\frac{\partial C}{\partial x}$ can be used to close the species transport equation. However, when there are multiple (N>2) components in the system, where each specie has different gradient and diffusivity parameter, the flux may not follow the Fick law.

The Maxwell–Stefan model captures this intrinsic coupling by balancing driving forces with pairwise friction.

$$\nabla y_s = \sum_{c=1}^{N} B_{sc}\, y_s y_c\, (\mathbf{u}_c - \mathbf{u}_s) = \frac{1}{n}\sum_{c=1}^{N} B_{sc}\, (y_s \mathbf{k}_c - y_c \mathbf{k}_s), \tag{3.5}$$

where B_{sc} are Maxwell–Stefan *resistance* coefficients and $\mathbf{k}_s$ denotes the molar diffusion flux. Equation (3.5) makes the coupling explicit: the gradient of one species depends on the relative motion of all species. Moreover, the diffusive flux may affect flow behavior and create advection [4].

3.3. Discrete MS–LBM formulation

3.3.1. D2Q9 lattice and hydrodynamic moments

In a standard D2Q9 LBM of 2D flow, the velocity at each lattice node is represented by 9 components, each represent the (mass) density of the flow in its direction. The 0th component is at rest, and the components 1~9 are the flow direction to the 9 adjacent nodes. We employ the standard D2Q9 lattice with $\Delta x = \Delta t = 1$ and sound speed $c_s^2 = 1/3$ [8–10]. Using the clockwise ordering $i = 0$ (rest), then north, north-east, east, south-east, south, south-west, west, north-west,

$$\begin{aligned} c_x &= [0,\ 0,\ 1,\ 1,\ 1,\ 0,\ -1,\ -1,\ -1], \\ c_y &= [0,\ 1,\ 1,\ 0,\ -1,\ -1,\ -1,\ 0,\ 1], \end{aligned} \tag{3.6}$$

with weights

$$w = \left[\tfrac{4}{9},\ \tfrac{1}{9},\ \tfrac{1}{36},\ \tfrac{1}{9},\ \tfrac{1}{36},\ \tfrac{1}{9},\ \tfrac{1}{36},\ \tfrac{1}{9},\ \tfrac{1}{36}\right]. \tag{3.7}$$

For each species,

$$\rho_s = \sum_{i=0}^{8} f_{s,i}, \qquad q_{x,s} = \sum_{i=0}^{8} c_{x,i} f_{s,i}, \qquad q_{y,s} = \sum_{i=0}^{8} c_{y,i} f_{s,i}, \tag{3.8}$$

and $u_s = q_s/\rho_s$.

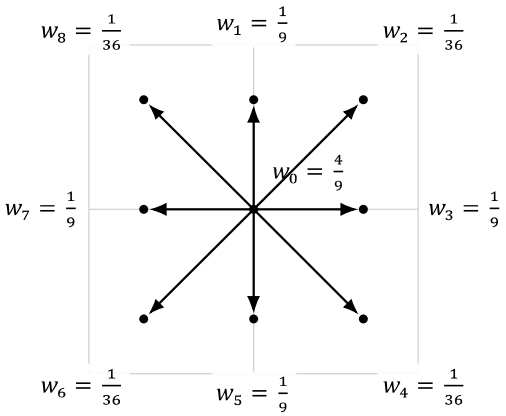


Figure 3.1: D2Q9 stencil with velocity index i and weight w_i (lattice units).

3.3.2. Equation of state and molecular-weight scaling

We define a mixture pressure through parameters ϕ_s,

$$p_{\text{mix}} = \sum_{s=1}^{N} \frac{\phi_s}{3} \rho_s, \qquad c_{\text{mix}}^2 \equiv \frac{p_{\text{mix}}}{\rho_{\text{mix}}}. \tag{3.9}$$

$$\phi_s = \phi_1 \frac{M_1}{M_s}, \tag{3.10}$$

We also define the local mixture molecular weight

$$M_{\text{mix}} = \left(\sum_{s=1}^{N} \frac{Y_s}{M_s} \right)^{-1}. \tag{3.11}$$

3.3.3. Maxwell–Stefan resistance model and coupling coefficients

Binary Maxwell–Stefan resistances B_{sc} between species s and c is parameterized as

$$B_{sc} = \beta \left(\frac{1}{M_s} + \frac{1}{M_c} \right)^{-1/2}, \tag{3.12}$$

where M_i is the molar mass of the molecule.

Define the non-dimensional coupling weights (for $c \neq s$)

$$\chi_{sc} = \frac{M_{\text{mix}}^2}{M_s M_c} \frac{B_{sc}}{B_{ss}}, \qquad (c \neq s), \tag{3.13}$$

and the species relaxation strength

$$\lambda_s = c_{\text{mix}}^2 B_{ss} = \frac{p_{\text{mix}}}{\rho_{\text{mix}}} B_{ss}. \tag{3.14}$$

We will discuss the choice of the parameters in section 4.1.

3.3.4. Coupling-aware equilibrium via a modified velocity

In Maxwell–Stefan diffusion, interspecies friction couples the momenta. The equilibrium with a modified velocity is calculated from

$$\mathbf{u}_s^* = \mathbf{u}_s + \sum_{\substack{c=1 \\ c \neq s}}^{N} \chi_{sc} \, Y_c \, (\mathbf{u}_c - \mathbf{u}_s) \tag{3.15}$$

The D2Q9 equilibrium distributions are (for $i \neq 0$)

$$f_{s,i}^{eq} = w_i \rho_s \left(\phi_s + 3 \, \mathbf{c}_i \cdot \mathbf{u}_s^* + \frac{9}{2} \left(\mathbf{c}_i \cdot \mathbf{u}_s^* \right)^2 - \frac{3}{2} |\mathbf{u}_s^*|^2 \right), \tag{3.16}$$

and for $i = 0$ we adjust the constant term to enforce $\sum_i f_{s,i}^{eq} = \rho_s$:

$$f_{s,0}^{eq} = w_0 \rho_s \left(\frac{9 - 5\phi_s}{4} - \frac{3}{2} |\mathbf{u}_s^*|^2 \right). \tag{3.17}$$

In the implementation we clip $f_{s,i}^{eq} \geq 0$ to prevent negative populations [4].

3.3.5. Trapezoidal BGK collision in transformed variables

A direct forward-Euler BGK update [11] can be restrictive for large relaxation strengths. We therefore use the trapezoidal (Crank–Nicolson) discretization through the transformed distribution

$$g_{s,i} = f_{s,i} - \theta\lambda_s \left(f_{s,i}^{eq} - f_{s,i}\right), \qquad \theta = \frac{1}{2}. \tag{3.18}$$

In this transformation, the mass and velocity are conserved. In g-space the collision is explicit with effective relaxation rate

$$\omega_s^{\text{eff}} = \frac{\lambda_s}{1+\theta\lambda_s}, \tag{3.19}$$

and post-collision populations

$$g_{s,i}^{\dagger} = g_{s,i} - \omega_s^{\text{eff}} \left(g_{s,i} - f_{s,i}^{eq}\right). \tag{3.20}$$

Streaming and boundary conditions are then applied to $g^{\dagger}$. After boundary updates and Maxwell–Stefan flux correction (Section 3.4), we reconstruct [4, 6]

$$f_{s,i}^{n+1} = \frac{g_{s,i}^{n+1} + \theta\lambda_s^{n+1} f_{s,i}^{eq,n+1}}{1+\theta\lambda_s^{n+1}}, \tag{3.21}$$

and clip to enforce non-negativity.

3.3.6. Streaming

After the trapezoidal BGK collision in transformed variables (Eq. (3.20)), the post-collision populations $g_{s,i}^{\dagger}(\mathbf{x}, t)$ are advected exactly along the discrete lattice velocities $\mathbf{c}_i$. With $\Delta x = \Delta t = 1$ (Sec. 3.3.1), the streaming step is the integer shift

$$g_{s,i}^{\text{str}}(\mathbf{x} + \mathbf{c}_i, t+1) = g_{s,i}^{\dagger}(\mathbf{x}, t), \qquad i \in \{0, \dots, 8\}, \tag{3.22}$$

equivalently, in the commonly used *pull* form,

$$g_{s,i}^{\text{str}}(\mathbf{x}, t+1) = g_{s,i}^{\dagger}(\mathbf{x} - \mathbf{c}_i, t). \tag{3.23}$$

The rest population ($i = 0$, $\mathbf{c}_0 = \mathbf{0}$) remains at the same node, while each moving population ($i \neq 0$) shifts to its neighboring lattice node in direction $\mathbf{c}_i$. Streaming is performed independently for each species; interspecies coupling is not applied during advection but is enforced (i) through the coupling-aware equilibrium (Sec. 3.3.4) and (ii) through the post-streaming Maxwell–Stefan flux correction (Sec. 3.4).

At physical boundaries, some streamed populations would originate outside the computational domain. After performing (3.23) we therefore overwrite the unknown incoming populations using the boundary rules in Sec. 3.5. For consistency with Sec. 3.4, we denote the populations *after streaming and boundary updates* by $g_{s,i}^{+}(\mathbf{x})$, which are then used to evaluate $\rho_s^{+} = \langle g_s^{+} \rangle$ and the streamed first moments $m_{s\alpha} = \langle c_\alpha g_s^{+} \rangle$.

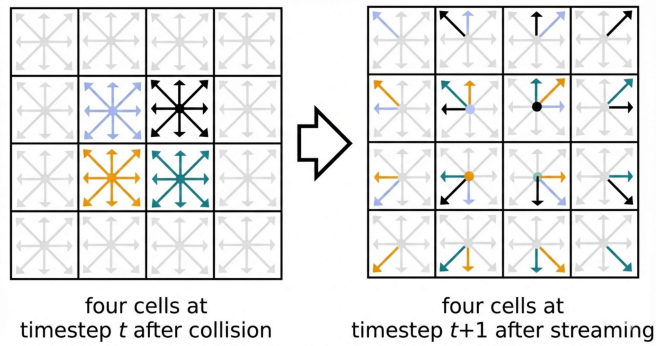


Figure 3.2: Illustration of the D2Q9 streaming step on a local 4×4 patch. Left: four neighboring lattice nodes at time t after collision, each carrying nine populations (eight moving directions plus the rest population at the center). Right: at time $t+1$ after streaming, every moving population has shifted by one lattice spacing along its discrete velocity, while the rest population remains at its original node. Colors track populations by their node of origin.

3.4. Post-streaming Maxwell–Stefan flux correction

After trapezoidal BGK collision in transformed variables (Section 3.3.5), streaming and boundary updates are applied, yielding $g_{s,i}^{+}(\mathbf{x})$, we calculate the Maxwell-Stefan flux correction to the multi-species flow. The transformation (3.18) preserves the zeroth moment, the post-step densities are obtained directly as

$$\rho_s^{+}(\mathbf{x}) = \langle g_s^{+} \rangle \equiv \sum_{i=0}^{8} g_{s,i}^{+}(\mathbf{x}). \tag{3.24}$$

The first moment of g_s^{+} is not the physical species momentum, since Maxwell–Stefan (MS) friction is embedded semi-implicitly in the θ-scheme. We therefore define the streamed first moments

$$m_{s\alpha}(\mathbf{x}) \equiv \langle c_\alpha g_s^{+} \rangle = \sum_{i=0}^{8} c_{\alpha,i}\, g_{s,i}^{+}(\mathbf{x}), \qquad \alpha \in \{x, y\}, \tag{3.25}$$

and introduce the unknown (MS-consistent) post-step species momenta

$$q_{s\alpha}^{+}(\mathbf{x}) \equiv \rho_s^{+}(\mathbf{x})\, u_{s\alpha}^{+}(\mathbf{x}). \tag{3.26}$$

Let $Y_s^{+} = \rho_s^{+}/\rho_{\mathrm{mix}}^{+}$ denote the local mass concentration with $\rho_{\mathrm{mix}}^{+} = \sum_s \rho_s^{+}$. Using the coupling weights $\chi_{sc}(\mathbf{x})$ and relaxation strengths $\lambda_s(\mathbf{x})$ (Eqs. (3.13)–(3.14)), define the projected coupling sum

$$\mathrm{X}_s^{+}(\mathbf{x}) \equiv \sum_{c=1}^{N} \chi_{sc}(\mathbf{x})\, Y_c^{+}(\mathbf{x}). \tag{3.27}$$

Taking the first moment of the transformed MS–BGK update and collecting unknown fluxes yields, for each component $\alpha \in \{x, y\}$, the local $N \times N$ system [4, 6, 12]

$$m_{s\alpha}(\mathbf{x}) = \Big[1 + \theta\, \lambda_s(\mathbf{x})\, \mathrm{X}_s^{+}(\mathbf{x})\Big]\, q_{s\alpha}^{+}(\mathbf{x}) - \theta\, \lambda_s(\mathbf{x})\, Y_s^{+}(\mathbf{x}) \sum_{c=1}^{N} \chi_{sc}(\mathbf{x})\, q_{c\alpha}^{+}(\mathbf{x}), \qquad s = 1, \ldots, N. \tag{3.28}$$

Equation (3.28) is solved *independently at each lattice node*. In matrix form,

$$\mathbf{A}(\mathbf{x})\, \mathbf{q}_\alpha^{+}(\mathbf{x}) = \mathbf{m}_\alpha(\mathbf{x}), \qquad \alpha \in \{x, y\}, \tag{3.29}$$

where $\mathbf{q}_\alpha^{+} = (q_{1\alpha}^{+}, \ldots, q_{N\alpha}^{+})^{\mathsf{T}}$ and $\mathbf{m}_\alpha = (m_{1\alpha}, \ldots, m_{N\alpha})^{\mathsf{T}}$. The coefficients are

$$A_{sc}(\mathbf{x}) = \delta_{sc}\Big[1 + \theta\, \lambda_s(\mathbf{x})\, \mathrm{X}_s^{+}(\mathbf{x})\Big] - \theta\, \lambda_s(\mathbf{x})\, Y_s^{+}(\mathbf{x})\, \chi_{sc}(\mathbf{x}). \tag{3.30}$$

Solving (3.29) provides the MS-consistent species momenta and velocities,

$$u_{s\alpha}^{+}(\mathbf{x}) = \frac{q_{s\alpha}^{+}(\mathbf{x})}{\rho_s^{+}(\mathbf{x})}, \qquad \alpha \in \{x, y\}, \tag{3.31}$$

(with $u_{s\alpha}^{+} = 0$ when $\rho_s^{+} = 0$). These corrected velocities are then used to rebuild the coupling-aware equilibrium (Section 3.3.4) and to perform the final back-transform (3.21).

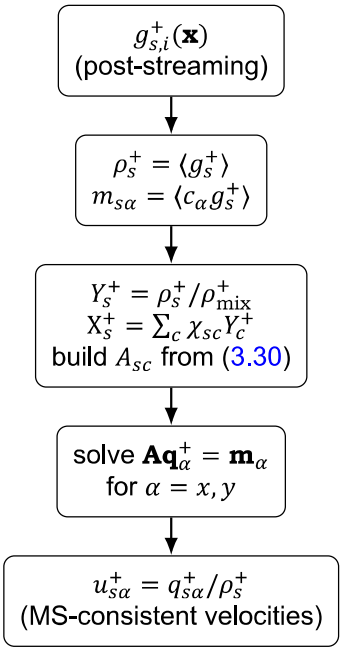


Figure 3.3: Local Maxwell–Stefan flux correction. Streamed g-moments provide $\mathbf{m}_\alpha$, while interspecies friction enters through $\mathbf{A}$, yielding corrected momenta $\mathbf{q}^+_\alpha$ and velocities $\mathbf{u}^+_s$.

3.5. Boundary conditions in transformed variables

All boundary conditions are imposed after streaming in the transformed-population space $g_{s,i}$. Within each time step we compute $g^\dagger$ by collision, stream to obtain g^{str}, and then overwrite the unknown incoming populations on boundaries to form g^{n+1}. This g^{n+1} is used for moment evaluation and Maxwell–Stefan correction before reconstructing f (Eq. (3.21)).

3.5.1. Unknown populations and opposite directions

At a boundary we update those populations that would stream *into* the domain from outside. For D2Q9 we denote by $\bar{i}$ the opposite direction of i. The incoming index sets used below are:

$$\mathcal{I}_{\mathrm{in}} = \{2,3,4\}, \qquad \mathcal{I}_{\mathrm{bot}} = \{1,2,8\}, \qquad \mathcal{I}_{\mathrm{top}} = \{4,5,6\}, \qquad \mathcal{I}_{\mathrm{out}} = \{6,7,8\}. \tag{3.32}$$

3.5.2. Left inlet: prescribed composition and velocity profile

The flow enters region of simulation from the left boundary, with prescribed velocity and concentration. The inlet velocity profile is given by the laminar flow solution: Poiseuille flow for the reactor-scale simulation, and shear flow for the $\approx 500um$ region on top of the wafer.

- **Couette flow inlet:**

$$u_x(y) = u_{\mathrm{wall}} \frac{y}{H}, \qquad H = n_y - 1. \tag{3.33}$$

- **Poiseuille flow inlet:**

$$u_x(y) = 4u_{\max}\,\eta(1-\eta), \qquad \eta = \frac{y}{H}, \quad H = n_y - 1. \tag{3.34}$$

We implemented the following inlet boundary conditions and found out only the modified Zou/He method works without leading to unrealistic behavior.

(i) Enforced to equilibrium distribution The most trivial way is to enforce the populations $f_i(x=0)$ to be the equilibrium distribution from equation 3.16 and 3.17. In this method, populations at $x = 0$ are

overwritten by the local equilibrium evaluated at $(\rho_{s,\mathrm{in}}, \mathbf{u}_{\mathrm{in}}(y))$:

$$g_{s,i}(0,y) \leftarrow f_{s,i}^{eq}\big(\rho_{s,\mathrm{in}}, \mathbf{u}_{\mathrm{in}}(y)\big), \qquad \forall i \in \{0,\dots,8\}. \tag{3.35}$$

In practice, this method overlooks the difference between the populations going leftwards (outwards). When the flow initialized with the same density as the inlet, the inlet doesn't provide enough pressure that drive the flow moving rightwards. This resulting in unrealistic backflow, as shown in supplement figure A.1.

(ii) Non-equilibrium extrapolation (NEE) inlet At the left boundary ($x = 0$) we prescribe the inflow composition $C_s^{\mathrm{in}}(y)$ together with a target advective velocity profile $\mathbf{u}^{\mathrm{in}}(y)$ (Poiseuille profile for the tube-flow test case). Rather than resetting all populations to equilibrium, this method apply a *non-equilibrium extrapolation* (NEE) inlet, which only reconstructs the populations that stream *into* the domain.

After streaming, the unknown populations at the inlet are those whose discrete velocities point from the boundary into the domain ($c_{x,i} > 0$). For each species s, we write the boundary population as an equilibrium part built from the prescribed inlet macros plus a non-equilibrium correction borrowed from the first interior column ($x = 1$):

$$f_{s,i}(x{=}0,y) = f_{s,i}^{\mathrm{eq}}\big(C_s^{\mathrm{in}}(y), \mathbf{u}^{\mathrm{in}}(y)\big) + \Big[f_{s,i}(x{=}1,y) - f_{s,i}^{\mathrm{eq}}\big(C_s(x{=}1,y), \mathbf{u}^{\mathrm{in}}(y)\big)\Big], \qquad c_{x,i} > 0. \tag{3.36}$$

Here $f_{s,i}^{\mathrm{eq}}$ is the species equilibrium distribution (Sec. 3.3), and the bracketed term represents the interior non-equilibrium contribution, which carries viscous/gradient information and reduces spurious reflections compared with a full equilibrium reset.

However, this method does not enforce prescribed velocity or momentum at the inlet. Eq. (3.36) is applied only to the *incoming* populations ($c_{x,i} > 0$); the remaining populations ($c_{x,i} \le 0$) are left unchanged and are determined by the interior solution. Consequently, the macroscopic moments evaluated at the inlet node,

$$C_s(x{=}0,y) = \sum_i f_{s,i}(x{=}0,y), \qquad \mathbf{j}_s(x{=}0,y) = \sum_i \mathbf{c}_i f_{s,i}(x{=}0,y), \tag{3.37}$$

contain contributions from both reconstructed incoming and untouched outgoing populations. Therefore, the node-wise velocity (or any momentum-like quantity derived from first moments) is *not* exactly constrained to $\mathbf{u}^{\mathrm{in}}$. Instead, the inlet condition prescribes the *injected* advective flux through the incoming populations. As a result, we observed backflow at the inlet, as shown in supplementary figure A.2.

(iii) Zou/He velocity inlet Zou/He velocity inlet method [13] calculates the $f_i(x = 0)$ based on prescribed velocity profile and density that calculated based on known components density. For a boundary node at the left inlet, the unknown components are 2, 3 and 4 (figure 3.3.1). It considers the bounce-back terms for the non-equilibrium part.

$$f_{i,\mathrm{unknown}} - f_{i,\mathrm{unknown}}^{eq}(u_x) = f_{\bar{i}} - f_{\bar{i}}^{eq}(u_x) \tag{3.38}$$

For an unknown component i, the non-equilibrium part of its opposite component $\bar{i}$ bounces back, and therefore the difference between this component and its equilibrium distribution f_i^{eq} at the designated velocity u_x is equal to that of the opposite component.

To calculate the total density at the boundary node, we first calculate the sum of the unknown components. Here $f_i = \sum_s f_{s,i}$ represent the density of component i of all the species s.

$$\sum_{i\in[2,3,4]} f_i = \sum_{\bar{i}\in[6,7,8]} f_{\bar{i}} + \sum_{i\in[2,3,4]} \left(f_i^{eq}(\rho_{tot}, u_x) - f_{\bar{i}}^{eq}(\rho_{tot}, u_x)\right) \tag{3.39}$$

where the total density ρ_{tot} is the sum of unknown and known components.

$$\begin{aligned}\rho_{tot} &= \underbrace{\sum_{i\in[2,3,4]} f_i}_{unknown} + \underbrace{\sum_{i\in[0,1,5,6,7,8]} f_i}_{known} \\ &= \sum_{\bar{i}\in[6,7,8]} f_{\bar{i}} + \rho_{tot} \sum_{i\in[2,3,4]} (B_i(u_x) - B_{\bar{i}}(u_x)) + \sum_{i\in[0,1,5,6,7,8]} f_i\end{aligned} \tag{3.40}$$

where $B(u_x)$ is the equilibrium state of density $= 1$ and velocity u_x. The total density ρ_{tot} can be expressed as

$$\rho_{tot} = \frac{\sum_{\bar{i}\in[6,7,8]} f_{\bar{i}} + \sum_{i\in[0,1,5,6,7,8]} f_i}{1 - \sum_{i\in[2,3,4]} (B_i(u_x) - B_{\bar{i}}(u_x))} \tag{3.41}$$

In Zou/He method, the density of each component can not be enforced as prescribed like the other two methods. Instead, the fraction of each component can be applied to total density to calculate their density.

$$\rho_s = x_s \rho_{tot} \tag{3.42}$$

where x_s denotes the molar fraction of the component s.

Finally, the incoming unknown populations can be reconstructed by

$$f_{s,i} = f_{s,i}^{eq} + (f_{s,\bar{i}} - f_{s,\bar{i}}^{eq}) \tag{3.43}$$

In our test, **this is the only method that worked** and did not shown unrealistic behavior, especially backflow.

3.5.3. Bottom wall: partially reactive boundary with masking

The bottom boundary ($y = 0$) represents a wafer segment that consumes TMIn and PH_3 and produces CH_4.

$$\mathrm{In(CH_3)_3 + PH_3 \rightarrow InP(s) + 3\,CH_4(g)}$$

(i) Carrier gas H_2: impermeable bounce-back. For the carrier gas (species $\mathrm{H_2}$) we enforce

$$g_{\mathrm{H_2},i}(x, 0) \leftarrow g_{\mathrm{H_2},\bar{i}}(x, 0), \qquad i \in \mathcal{I}_{\mathrm{bot}}. \tag{3.44}$$

(ii) TMIn: Robin condition. For TMIn we impose a linear Robin boundary condition on the wall concentration C_s,

$$b_1\, \partial_n C_s + b_2\, C_s = b_3 \qquad \text{on } y = 0, \tag{3.45}$$

where ∂_n is the inward normal derivative (into the domain, $+y$). When $b_3 = 0$,

$$\frac{\partial c}{\partial y} = -\frac{b_2}{b_1} c \tag{3.46}$$

which is the same boundary condition used in VPD model in section 2.2.3 equation 2.4. The ”effective diffusion length” D/k in VPD model is $\frac{b_2}{b_1}$.

Using a half-cell finite difference,

$$\partial_n C_s \approx \frac{C_{f,s} - C_{w,s}}{\Delta x/2}, \tag{3.47}$$

and solving (3.45) for the effective wall value $C_{w,s}$ gives

$$C_{w,s} = \frac{C_{f,s} + \frac{\Delta x}{2}\frac{b_3}{b_1}}{1 + \frac{\Delta x}{2}\frac{b_2}{b_1}}. \tag{3.48}$$

On the reacting segment we impose $C_{w,s}$ through a half-way anti-bounce-back update

$$g_{s,i}(x,0) \leftarrow -g_{s,\bar{i}}(x,0) + 2f_{s,i}^{eq}\big(C_{w,s}, \mathbf{u}_w = \mathbf{0}\big), \qquad i \in \mathcal{I}_{\text{bot}}. \tag{3.49}$$

On the non-reacting segment (mask $= 1$) we revert to impermeable bounce-back $g_{s,i} \leftarrow g_{s,\bar{i}}$ [14, 15].

(iii) Stoichiometric coupling: PH_3 and CH_4. We assume the overall surface reaction

$$\mathrm{In(CH_3)_3} + \mathrm{PH_3} \rightarrow \mathrm{InP} + 3\,\mathrm{CH_4}, \tag{3.50}$$

so the wall fluxes satisfy

$$J_{\mathrm{PH_3}} = +1 \cdot J_{\mathrm{TMIn}}, \qquad J_{\mathrm{CH_4}} = -3 \cdot J_{\mathrm{TMIn}}. \tag{3.51}$$

We impose these as Neumann (flux) conditions by converting J_s into an effective wall value using

$$C_{w,s} = C_{f,s} - \frac{\Delta x}{2}\frac{J_s}{D_s}, \tag{3.52}$$

and then applying the same anti-bounce-back update (3.49). A negative J_s (CH_4 production) yields $C_{w,s} > C_{f,s}$ and injects CH_4 into the gas phase.

There are two ways to calculate J_{TMIn}: (i) based on concentration of TMIn at the boundary, and use use the theoretical value to calculate the flux. (ii) based on the concentration gradient of TMIn in a few grids close to the boundary. The former one is inaccurate in diffusion-limited case, but very accurate otherwise; the latter one is less accurate in general, but more accurate in diffusion-limited case.

3.5.4. Right outlet: zero-gradient outflow

At the right boundary ($x = n_x - 1$) we apply a simple zero-gradient extrapolation for incoming populations:

$$g_{s,i}(n_x{-}1, y) \leftarrow g_{s,i}(n_x{-}2, y), \qquad i \in \mathcal{I}_{\text{out}}. \tag{3.53}$$

This approximates $\partial_x g_{s,i} = 0$ at the outlet and was sufficient for the laminar reactor-scale cases considered here.

3.5.5. Top boundary

No-slip wall For the reactor scale simulation, we impose no-slip using a half-way bounce-back rule. For a stationary wall ($\mathbf{u}_w = \mathbf{0}$),

$$g_{s,i}(x, n_y{-}1) \leftarrow g_{s,\bar{i}}(x, n_y{-}1), \qquad i \in \mathcal{I}_{\text{top}}. \tag{3.54}$$

Moving boundary with prescribed concentration At the moving top wall we initially imposed a Dirichlet condition for the species distributions using an anti–bounce-back reconstruction,

$$g_i(\mathbf{x}_b) = -\,g_{\bar{i}}(\mathbf{x}_b) + 2\,g_i^{\mathrm{eq}}(C_w, \mathbf{u}_w)\,, \qquad i \in \mathcal{I}_{\text{in}}, \tag{3.55}$$

where $\mathcal{I}_{\text{in}}$ denotes the set of incoming directions at the boundary node $\mathbf{x}_b$, $\bar{i}$ is the opposite lattice direction, and $g_i^{\mathrm{eq}}(C_w, \mathbf{u}_w)$ is the local equilibrium built with prescribed concentration C_w and wall velocity $\mathbf{u}_w$. Decomposing $g_{\bar{i}} = g_{\bar{i}}^{\mathrm{eq}} + g_{\bar{i}}^{\mathrm{neq}}$ shows that this rule injects an amplified odd (momentum-carrying) part of the equilibrium:

$$g_i - g_{\bar{i}} = 2\left(g_i^{\mathrm{eq}} - g_{\bar{i}}^{\mathrm{eq}}\right) - 2g_{\bar{i}}^{\mathrm{neq}}, \tag{3.56}$$

so the boundary momentum contribution of each opposite pair is doubled relative to the target equilibrium and additionally contaminated by the interior non-equilibrium term. This explains the observed over-injection of tangential velocity at the lid when $\mathbf{u}_w \neq \mathbf{0}$. To prescribe both C_w and $\mathbf{u}_w$ without amplifying momentum, we instead use a non-equilibrium extrapolation (NEE) reconstruction,

$$g_i(\mathbf{x}_b) = g_{\bar{i}}(\mathbf{x}_b) + g_i^{\mathrm{eq}}(C_w, \mathbf{u}_w) - g_{\bar{i}}^{\mathrm{eq}}(C_w, \mathbf{u}_w)\,, \tag{3.57}$$

which enforces $g_i - g_i^{\mathrm{eq}} = g_{\bar{i}} - g_{\bar{i}}^{\mathrm{eq}}$ and cancels the non-equilibrium contribution in the pair momentum, yielding the intended boundary forcing.

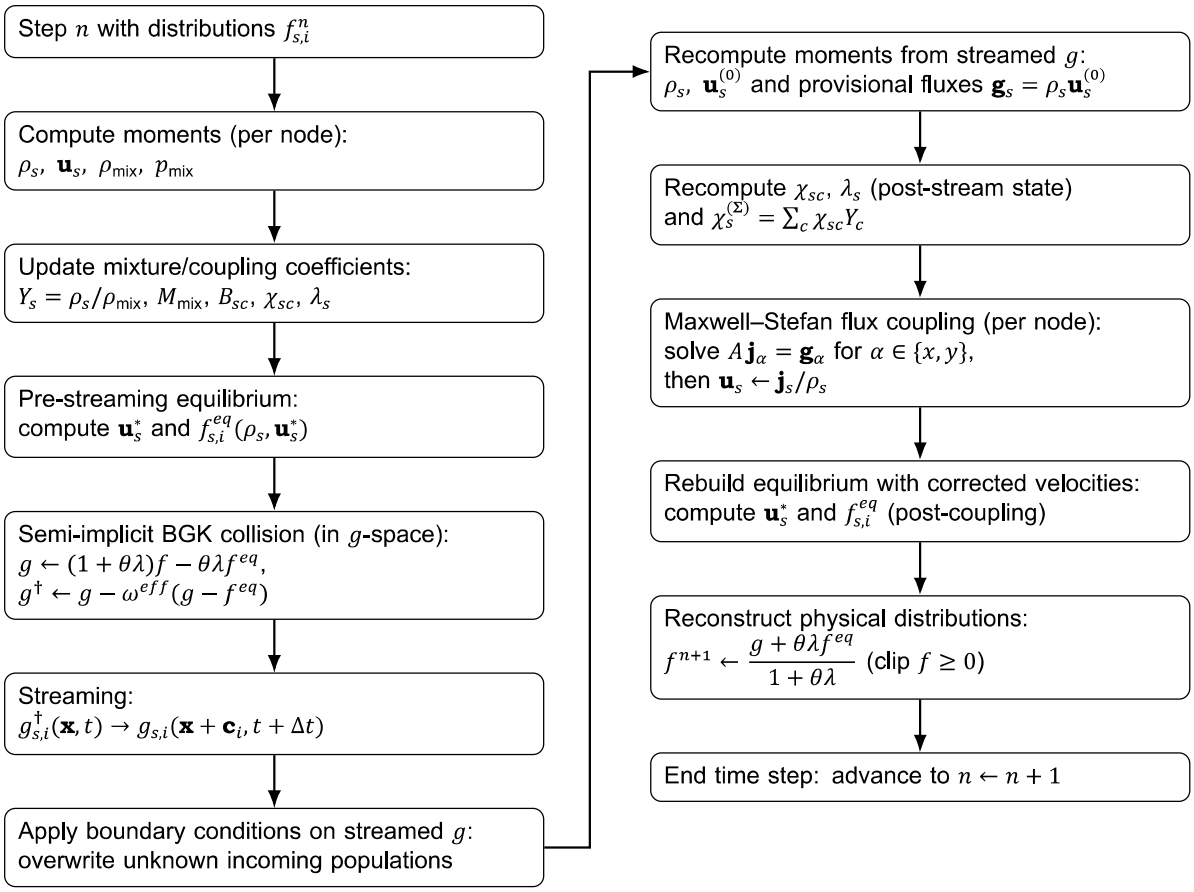


Figure 3.4: Workflow of one MS–LBM time step: trapezoidal BGK collision in transformed variables, streaming and boundary updates in g-space, per-node Maxwell–Stefan momentum correction, and reconstruction of f.

4

Simulations and Results

4.1. Matching transport regimes with dimensionless numbers

To reproduce the transport behavior of the MOCVD reactor with MS–LBM, we match the relevant dimensionless groups between the physical system and the lattice model. The key controls are the Reynolds number Re (inertia vs. viscosity) and the Péclet number Pe_s (advection vs. diffusion) for limiting species (e.g. TMIn in InP growth). From these we also obtain the Schmidt number Sc_s (momentum diffusion rate by viscosity versus mass diffusion rate) and, when needed, external mass-transfer estimates via the Sherwood number Sh_s.

We perform the simulation in multiscale, here we discuss dimensionless numbers at two characteristic length scales: (i) the part of the reactor where a 38mm wafer is located (”reactor scale”), and (ii) a near-surface micro-layer (hundreds of μm) where the transport-limited reactant supply affect SAG result.

4.1.1. Definitions

For characteristic length L and velocity scale U, the standard groups are

$$\mathrm{Re} = \frac{\rho U L}{\mu_{\mathrm{dyn}}} = \frac{UL}{\nu}, \tag{4.1}$$

$$\mathrm{Sc}_s = \frac{\nu}{D_s} = \frac{\mu_{\mathrm{dyn}}}{\rho D_s}, \tag{4.2}$$

$$\mathrm{Pe}_s = \frac{UL}{D_s} = \mathrm{Re}\,\mathrm{Sc}_s, \tag{4.3}$$

where ρ is gas density, μ_{dyn} the dynamic viscosity, $\nu = \mu_{\mathrm{dyn}}/\rho$ the kinematic viscosity, and D_s the molecular diffusion coefficient of species s in the carrier gas (H_2).

To estimate external mass transfer, we define the Sherwood number

$$\mathrm{Sh}_s = \frac{k_{m,s} L}{D_s}, \tag{4.4}$$

and for laminar forced convection over a flat plate we use the classical correlation (order-of-magnitude estimate)

$$\mathrm{Sh}_{L,s} \approx 0.664\,\mathrm{Re}_L^{1/2}\,\mathrm{Sc}_s^{1/3}, \qquad k_{m,s} \approx \frac{\mathrm{Sh}_{L,s} D_s}{L}. \tag{4.5}$$

The laminar boundary-layer thickness scaling is

$$\delta_{99}(x) \approx 5\,\frac{x}{\sqrt{\mathrm{Re}_x}}, \qquad \mathrm{Re}_x = \frac{\rho U x}{\mu_{\mathrm{dyn}}}. \tag{4.6}$$

In confined reactor gaps, δ_{99} (boundary layer thickness: the point where the velocity of is 99% of free stream) cannot exceed the local gap height; therefore Eq. (4.6) is used here as a consistency check rather than a strict prediction.

Finally, to compare surface incorporation to gas-phase transport we define a Damköhler-type ratio

$$\mathrm{Da} \equiv \frac{k_{\mathrm{rxn}}}{k_{m,s}}, \tag{4.7}$$

where k_{rxn} is an effective first-order surface rate constant (m/s).

4.1.2. Flow-rate conversion and gas properties

The flow rate is given as a standard volumetric flow $\dot{V}_{\mathrm{std}}$ (sccm/slm). Assuming ideal-gas behavior, the actual volumetric flow rate inside the reactor is

$$\dot{V} = \dot{V}_{\mathrm{std}} \left(\frac{T}{T_{\mathrm{std}}} \right) \left(\frac{p_{\mathrm{std}}}{p} \right), \tag{4.8}$$

with standard conditions $(T_{\mathrm{std}}, p_{\mathrm{std}})$ and reactor conditions (T, p).

For the InP run:

- Total standard flow: $\dot{V}_{\mathrm{std}} = 15.3\ \mathrm{slm} = 15300\ \mathrm{sccm}$,
- Reactor temperature: $T = 650^{\circ}\mathrm{C} = 923.15\ \mathrm{K}$,
- Reactor pressure: $p = 100\ \mathrm{mbar}$,
- Standard conditions: $(T_{\mathrm{std}}, p_{\mathrm{std}}) = (298.15\ \mathrm{K},\ 1013.25\ \mathrm{mbar})$.

This gives

$$\dot{V} \approx 8.00 \times 10^{-3}\ \mathrm{m}^3/\mathrm{s}. \tag{4.9}$$

We estimate the (H_2-dominated) density from the ideal gas law

$$\rho \approx \frac{p M_{\mathrm{H_2}}}{RT}, \tag{4.10}$$

with $M_{\mathrm{H_2}} = 2.016 \times 10^{-3}\ \mathrm{kg/mol}$. At $p = 10^4\ \mathrm{Pa}$ and $T = 923.15\ \mathrm{K}$ this gives

$$\rho \approx 2.63 \times 10^{-3}\ \mathrm{kg/m}^3. \tag{4.11}$$

For the dynamic viscosity we use the data of hydrogen gas from Hanley et.al. [16] $\mu_{\mathrm{H_2}} \approx 1.92 \times 10^{-5}\ \mathrm{Pa \cdot s}$, hence $\nu = \mu_{\mathrm{dyn}}/\rho \approx 7.31 \times 10^{-3}\ \mathrm{m}^2/\mathrm{s}$.

4.1.3. Diffusion coefficient estimation

To estimate gas-phase molecular diffusion coefficients, we use the Chapman–Enskog theory for dilute binary mixtures based on the Lennard-Jones (12-6) potential [17],

$$\varphi_{AB}(r) = 4\varepsilon_{AB} \left[\left(\frac{\sigma_{AB}}{r} \right)^{12} - \left(\frac{\sigma_{AB}}{r} \right)^{6} \right], \tag{4.12}$$

where σ_{AB} is the collision diameter and ε_{AB} is the well depth.

For ideal gases, the working equation for the binary diffusivity is

$$D_{AB} = 0.0018583\ \frac{T^{3/2}}{p\, \sigma_{AB}^2\, \Omega_{D,AB}(T_R)} \sqrt{\frac{1}{M_A} + \frac{1}{M_B}} \qquad [\mathrm{cm}^2/\mathrm{s}], \tag{4.13}$$

Table 4.1: Lennard–Jones transport parameters used in the Chapman–Enskog estimate.

Species	ε/k_B (K)	σ (Å)
H_2	38.0	2.920
PH_3	251.5	3.981
TMG ($GaMe_3$, surrogate for TMIn σ, ε)	378.2	5.52

with T in K, p in atm, M_i in g/mol, and σ_{AB} in Å. The Lennard–Jones mixing rules are

$$\sigma_{AB} = \frac{\sigma_A + \sigma_B}{2}, \qquad \epsilon_{AB} = \sqrt{\epsilon_A \epsilon_B}, \qquad T_R = \frac{T}{\varepsilon_{AB}/k_B}. \tag{4.14}$$

And $\Omega_{D,AB}(T_R)$ is the diffusion collision integral for diffusion, a function of T_R [18].

Because H_2 is the dominant carrier gas in the reactor, and PH_3 and TMIn are dilute, we approximate the mixture-averaged diffusivities by the binary values in H_2,

$$D_s \approx D_{s,\mathrm{H_2}}. \tag{4.15}$$

The Lennard–Jones parameters $(\varepsilon/k_B, \sigma)$ are taken from standard transport databases. For PH_3 we use the tabulated PH_3 entry, and for H_2 we use the corresponding H_2 entry. Transport parameters for TMIn are not consistently tabulated in common datasets; therefore we use trimethylgallium (TMG, $GaMe_3$) as a size/chemistry surrogate for the collision parameters (while using the true TMIn molecular weight in Eq. (4.13)).

Evaluating Eq. (4.13) at the reactor conditions gives

$$D_{\mathrm{PH_3–H_2}} \approx 4.29 \times 10^{-3}\ \mathrm{m^2/s}, \tag{4.16}$$

$$D_{\mathrm{TMIn–H_2}} \approx 2.71 \times 10^{-3}\ \mathrm{m^2/s}. \tag{4.17}$$

The TMIn value should be interpreted as an order-of-magnitude estimate; the dominant sensitivity is the effective collision diameter ($D \propto \sigma_{AB}^{-2}$), so uncertainty in the surrogate σ directly maps into uncertainty in $D_{\mathrm{TMIn–H_2}}$.

We estimate the diffusion coefficient from Chapman-Enskog theory

$$D_{\mathrm{TMIn}} \approx 2.60 \times 10^{-3}\ \mathrm{m^2/s}, \tag{4.18}$$

$$D_{\mathrm{PH_3}} \approx 4.24 \times 10^{-3}\ \mathrm{m^2/s}. \tag{4.19}$$

Therefore the Schmidt numbers are

$$\mathrm{Sc_{TMIn}} \approx \frac{\nu}{D_{\mathrm{TMIn}}} \approx 2.81, \tag{4.20}$$

$$\mathrm{Sc_{PH_3}} \approx \frac{\nu}{D_{\mathrm{PH_3}}} \approx 1.72. \tag{4.21}$$

As the V/III ratio is 435, the TMIn is the limiting factor to the process, and the PH_3 is sufficient, we mainly use D_{TMIn} to characterize the diffusion.

4.1.4. Reactor scale estimation

The reactive gas volume above the susceptor is approximately $V = 1.23\ \mathrm{L} = 1.23 \times 10^{-3}\ \mathrm{m^3}$. Approximating the cross-section as a rectangle of width $w = 13\ \mathrm{cm}$ and height $h = 4.5\ \mathrm{cm}$ gives

$$A = wh = 5.85 \times 10^{-3}\ \mathrm{m^2}, \qquad U_{\mathrm{ch}} \approx \frac{\dot{V}}{A} \approx 1.37\ \mathrm{m/s}. \tag{4.22}$$

A characteristic axial length follows from $L_{\mathrm{ch}} \approx V/A \approx 0.210\ \mathrm{m}$. For duct flow we also report the hydraulic diameter

$$D_h = \frac{2wh}{w + h} \approx 6.69 \times 10^{-2}\ \mathrm{m}. \tag{4.23}$$

Using $L = D_h$,

$$\mathrm{Re}_{D_h} \approx \frac{\rho U_{\mathrm{ch}} D_h}{\mu_{\mathrm{dyn}}} \approx 12.5, \tag{4.24}$$

indicating laminar flow at the macro scale. The corresponding Péclet number is

$$\mathrm{Pe}_{D_h,\mathrm{TMIn}} \approx \frac{U_{\mathrm{ch}} D_h}{D_{\mathrm{TMIn}}} \approx 35.2 \tag{4.25}$$

The chamber residence time is

$$\tau_{\mathrm{ch}} \approx \frac{V}{\dot{V}} \approx 0.154 \text{ s}. \tag{4.26}$$

The Péclet numbers much large than 1, indicating the advection is dominant at reactor scale.

If we zoom in to the length of the 38 mm wafer, take the characteristic length as the wafer/growth-zone length along the flow,

$$L_w = 38 \text{ mm} = 3.8 \times 10^{-2} \text{ m}, \tag{4.27}$$

and use a representative local velocity $U_w \approx 1$ m/s in the near-wafer region.

$$\mathrm{Re}_w \approx \frac{\rho U_w L_w}{\mu_{\mathrm{dyn}}} \approx 5.20, \tag{4.28}$$

$$\mathrm{Pe}_{w,\mathrm{TMIn}} \approx \frac{U_w L_w}{D_{\mathrm{TMIn}}} \approx 14.6 \tag{4.29}$$

Using Eq. (4.5) as an external-transfer estimate yields

$$\mathrm{Sh}_{w,\mathrm{TMIn}} \approx 2.14, \qquad k_{m,\mathrm{TMIn}} \approx 1.46 \times 10^{-1} \text{ m/s} \tag{4.30}$$

The Blasius scaling predicts $\delta_{99}(L_w) \sim 8.3$ cm, which exceeds typical reactor gaps; thus the near-wafer boundary layer is expected to be gap-limited rather than an unconfined flat-plate layer.

The measured growth rate is $\dot{h} = 180$ Å/min $= 3.0 \times 10^{-10}$ m/s. The incorporation flux of InP is estimated as

$$J_{\mathrm{InP}} \approx \frac{\dot{h}\,\rho_{\mathrm{InP}}}{M_{\mathrm{InP}}} \approx 9.90 \times 10^{-6} \text{ mol/(m}^2\text{ s)}, \tag{4.31}$$

using $\rho_{\mathrm{InP}} = 4.81 \times 10^3$ kg/m^3 and $M_{\mathrm{InP}} = 0.1458$ kg/mol. With TMIn partial pressure $p_{\mathrm{TMIn}} = 0.0469$ mbar $= 4.69$ Pa, the bulk TMIn concentration is $c_{\mathrm{TMIn}} = p_{\mathrm{TMIn}}/(RT) \approx 6.11 \times 10^{-4}$ mol/m^3, and an effective first-order surface rate (average molecule drift velocity) constant is

$$k_{\mathrm{rxn}} \approx \frac{J_{\mathrm{InP}}}{c_{\mathrm{TMIn}}} \approx 1.62 \times 10^{-2} \text{ m/s}. \tag{4.32}$$

The resulting Damköhler-type ratios (Eq. (4.7)) are

$$\mathrm{Da}_{\mathrm{TMIn}} \approx 0.11, \qquad \mathrm{Da}_{\mathrm{PH_3}} \approx 0.08, \tag{4.33}$$

suggesting that, under these assumptions, growth is not purely transport-limited at the wafer scale, but also partially limited by reaction rate.

4.1.5. Micro-layer scale estimation

To characterize a thin region of thickness L_μ close to the wafer, we use

$$L_\mu = 500\ \mu\text{m} = 5.0 \times 10^{-4} \text{ m}, \tag{4.34}$$

and define a velocity scale from a local shear rate $\dot{\gamma}$:

$$U_\mu \approx \dot{\gamma} L_\mu, \qquad \dot{\gamma} = 200 \text{ s}^{-1} \Rightarrow U_\mu \approx 0.10 \text{ m/s}. \tag{4.35}$$

This yields

$$\mathrm{Re}_\mu \approx \frac{\rho U_\mu L_\mu}{\mu_{\mathrm{dyn}}} \approx 6.84 \times 10^{-3}, \tag{4.36}$$

$$\mathrm{Pe}_{\mu,\mathrm{TMIn}} \approx \frac{U_\mu L_\mu}{D_{\mathrm{TMIn}}} \approx 1.92 \times 10^{-2} \tag{4.37}$$

Therefore, at $O(10^2\text{–}10^3\ \mu\mathrm{m})$ above the wafer, molecular diffusion dominates over advection ($\mathrm{Pe} \ll 1$), motivating careful treatment of surface reaction/absorption boundary conditions.

Table 4.2: Order-of-magnitude dimensionless numbers at three characteristic scales for the default InP run.

Scale	L (m)	U (m/s)	Re	$\mathrm{Sc}_{\mathrm{TMIn}}$	$\mathrm{Pe}_{\mathrm{TMIn}}$
Chamber	$L_{\mathrm{ch}} = 2.10 \times 10^{-1}$	1.37	39.3†	2.81	1.11×10^2
Wafer scale	$L_w = 3.8 \times 10^{-2}$	1.0	5.20	2.81	14.6
Micro-layer	$L_\mu = 5.0 \times 10^{-4}$	0.10	6.84×10^{-3}	2.81	1.92×10^{-2}

†This Reynolds number uses the axial length L_{ch} (not the conventional duct choice D_h), included only to form $\mathrm{Pe} = UL/D$ consistently on the same axial length.

4.2. Mapping physical transport to MS–LBM parameters

This section explains how we choose the MS–LBM parameters to match the target Pe and Re (or, equivalently, Pe and Sc) at a selected physical scale. In practice, we match TMIn most tightly because it is the rate-limiting precursor in the InP run.

4.2.1. LBM viscosity in terms of the semi-implicit relaxation

In the semi-implicit BGK update (Section 3.3.5), the lattice kinematic viscosity is

$$\nu_{\mathrm{LBM}} = c_s^2 \left(\frac{1}{\omega^{\mathrm{eff}}} - \frac{1}{2} \right), \qquad c_s^2 = \frac{1}{3}, \tag{4.38}$$

with $\omega^{\mathrm{eff}} = \lambda/(1+\lambda/2)$ (Eq. 3.19 with $\theta = \frac{1}{2}$). In the MS–LBM implementation, λ is defined by Eq. (3.14):

$$\lambda_s = c_{\mathrm{mix}}^2 B_{ss}, \qquad c_{\mathrm{mix}}^2 = \frac{p_{\mathrm{mix}}}{\rho_{\mathrm{mix}}}, \tag{4.39}$$

and for an H_2-dominated mixture one has $c_{\mathrm{mix}}^2 \approx \phi_{H_2}/3$ from Eq. (3.9). Therefore, using the H_2 self-resistivity as a dominant estimate,

$$\lambda \approx \left(\frac{\phi_{H_2}}{3} \right) B_{11}. \tag{4.40}$$

4.2.2. Diffusivity scaling

In our parameterization, the Maxwell–Stefan resistivities scale with molecular weights as

$$B_{s,H_2} = \beta\, \chi_s, \qquad \chi_s \equiv \left(\frac{1}{M_s} + \frac{1}{M_{H_2}} \right)^{-1/2}, \tag{4.41}$$

where β is a global multiplier and M_s are the species molecular weights. In lattice units, the diffusion coefficient scales as

$$D_s^{\mathrm{LBM}} \equiv D_{s,H_2}^{\mathrm{LBM}} \approx \frac{1}{B_{s,H_2}} = \frac{1}{\beta\, \chi_s}. \tag{4.42}$$

Thus, β primarily sets the overall magnitude of diffusivities, and therefore controls the lattice Péclet numbers. (Note: in principle β can be set to different values that for each species, allowing the diffusion coefficient to be fine-tuned based on molecule size, as discussed in section 4.1.3, but in our code we used the same β for all species for simplicity)

4.2.3. Matching dimensionless numbers

In the H_2-dominated regime, $\lambda \propto \beta\,\phi_{H_2}$ (Eq. (4.40)), while $D_s^{\mathrm{LBM}} \propto 1/\beta$ (Eq. (4.42)), so $\mathrm{Sc}_s^{\mathrm{LBM}} = \nu_{\mathrm{LBM}}/D_s^{\mathrm{LBM}}$ is primarily controlled by ϕ_{H_2} and weakly dependent on the exact λ regime.

$$\mathrm{Sc}_s^{\mathrm{LBM}} \approx \frac{\chi_s}{\phi_{H_2}\,\chi_{11}}, \qquad \chi_{11} \equiv \left(\frac{1}{M_{H_2}} + \frac{1}{M_{H_2}}\right)^{-1/2}, \tag{4.43}$$

and therefore the closed-form choice

$$\phi_{H_2} \approx \frac{\chi_s}{\chi_{11}\,\mathrm{Sc}_s^{\mathrm{phys}}}. \tag{4.44}$$

Once ϕ_{H_2} is set, we assign other ϕ_s using the molecular-weight scaling used in the code:

$$\phi_s = \phi_{H_2}\,\frac{M_{H_2}}{M_s}. \tag{4.45}$$

With ϕ of each species determined, the maximum velocity $u_{max,LBM}$ allowed in LBM can be determined. If the velocity exceed a certain value, a velocity populations may become negative, according to equations 3.16 and 3.17. To ensure the stability of the algorithm, the negative component is clipped to 0, at the cost of accuracy.

The next step is to find β that gives the desired diffusion coefficient. Imposing $\mathrm{Pe}_s^{\mathrm{LBM}} = \mathrm{Pe}_s^{\mathrm{phys}}$, and $u_{LBM} = u_{max,LBM}$ (or a value smaller than it) gives the required lattice diffusivity

$$D_{s,\mathrm{req}}^{\mathrm{LBM}} = \frac{u_{\mathrm{LBM}}\,N}{\mathrm{Pe}_s^{\mathrm{phys}}}. \tag{4.46}$$

Combining Eq. (4.46) with Eq. (4.42) yields

$$\beta = \frac{1}{\chi_s\,D_{s,\mathrm{req}}^{\mathrm{LBM}}} = \frac{\mathrm{Pe}_s^{\mathrm{phys}}}{u_{\mathrm{LBM}}\,N\,\chi_s}. \tag{4.47}$$

Finally, because $\mathrm{Pe} = \mathrm{Re}\,\mathrm{Sc}$ by definition, matching Pe and Sc implies that Re is matched automatically at the chosen scale:

$$\mathrm{Re}^{\mathrm{LBM}} = \frac{\mathrm{Pe}_s^{\mathrm{LBM}}}{\mathrm{Sc}_s^{\mathrm{LBM}}} \approx \frac{\mathrm{Pe}_s^{\mathrm{phys}}}{\mathrm{Sc}_s^{\mathrm{phys}}} = \mathrm{Re}^{\mathrm{phys}}. \tag{4.48}$$

4.2.4. Example: Calculating LBM parameters to match dimensionless numbers

This subsection makes the parameter-selection procedure in Section 4.2 concrete for the four-species InP case. The same MS–LBM transport parameters (ϕ, β) are used in both simulation scales (reactor scale, Section 4.3; micro-layer, Section 4.4); the two scales differ only in the characteristic lattice length N and velocity scale u_{LBM} used to realize the target dimensionless regimes (Table 4.2). The workflow for calculating these parameters is included in our codebase.

Step 1: specify molecular-weight ratios. Because only mass *ratios* enter the resistivity scaling (Eq. (4.41)) and the resulting dimensionless numbers, we work with the scaled molecular weights

$$M = \{M_{H_2}, M_{\mathrm{TMIn}}, M_{PH_3}, M_{CH_4}\} = \{1, 80, 17, 8\}.$$

Step 2: choose ϕ to match a target Schmidt number. In the H_2-dominated regime, Eq. (4.43) gives the closed-form estimate $\phi_{H_2} \approx \chi_s/(\chi_{11}\,\mathrm{Sc}_s^{\mathrm{phys}})$ (Eq. (4.44)). Using TMIn as the reference species and the reactor-scale values

$$\mathrm{Sc}_{\mathrm{TMIn}}^{\mathrm{phys}} = \frac{\mathrm{Pe}_{\mathrm{TMIn}}^{\mathrm{phys}}}{\mathrm{Re}^{\mathrm{phys}}} \approx 2.89,$$

gives

$$\phi_{H_2} \approx 0.487 \quad \text{(rounded to } 0.48 \text{ in the rest of this chapter).}$$

The remaining entries are assigned by the molecular-weight scaling used in the code (Eq. (4.45)),

$$\phi = \left[\phi_{H_2},\ \phi_{H_2}\frac{M_{H_2}}{M_{TMIn}},\ \phi_{H_2}\frac{M_{H_2}}{M_{PH_3}},\ \phi_{H_2}\frac{M_{H_2}}{M_{CH_4}}\right] \approx \left[4.87\times10^{-1},\ 6.09\times10^{-3},\ 2.86\times10^{-2},\ 6.09\times10^{-2}\right]. \tag{4.49}$$

Using the rounded value $\phi_{H_2} = 0.48$ instead gives $\phi = [0.48,\ 6.0\times10^{-3},\ 2.82\times10^{-2},\ 6.0\times10^{-2}]$.

Step 3: choose β for the reactor-scale simulation. Once a characteristic lattice velocity u_{LBM} and lattice length N are fixed, Eq. (4.46) and Eq. (4.47) give the value of β required to match a target Péclet number for a chosen reference species. In the multiscale workflow, ϕ is kept fixed because the Schmidt number is unchanged across scales; only β, u_{LBM}, and N are varied to match the target Péclet number. The Reynolds number then follows automatically because $\mathrm{Sc} = \mathrm{Pe}/\mathrm{Re}$ remains fixed.

For the reactor-scale simulation, we estimate the characteristic flow velocity at the center of the reactor to be $U \approx 1.5\ \mathrm{m/s}$ and take the characteristic length to be the wafer-scale transport distance $L = 3.8\times10^{-2}\ \mathrm{m}$. Together with the Chapman–Enskog estimate $D_{TMIn} \approx 2.6\times10^{-3}\ \mathrm{m^2/s}$ (Section 4.1.3), this gives

$$\mathrm{Pe}^{\mathrm{phys}}_{TMIn} = \frac{UL}{D_{TMIn}} \approx 21.9, \qquad \mathrm{Re}^{\mathrm{phys}} = \frac{\rho UL}{\mu} \approx 7.60, \tag{4.50}$$

and therefore $\mathrm{Sc}^{\mathrm{phys}} \approx 2.89$, consistent with Step 2.

In LBM, the 38 mm wafer-scale transport length is discretized with $N = 400$ lattice nodes. The admissible lattice velocity is limited by the equilibrium positivity bound and, for this mixture, by the heaviest species. From the code,

$$u_{max} = \min\left(v_{max}(\phi_{H_2}),\ v_{max}(\phi_{TMIn})\right) \approx 2.03\times10^{-3}\ \text{grid/step.}$$

To keep a safety margin from this stability limit, we use

$$u_{LBM} = 0.67\,u_{max} \approx 1.36\times10^{-3}\ \text{grid/step.}$$

The required lattice diffusivity of TMIn is then

$$D^{LBM}_{TMIn} = \frac{u_{LBM}\,N}{\mathrm{Pe}^{\mathrm{phys}}_{TMIn}} \approx 2.49\times10^{-2}\ \text{grid/step.} \tag{4.51}$$

Using Eq. (4.41) and Eq. (4.42), we obtain

$$\beta = \frac{1}{D^{LBM}_{TMIn}\chi_{TMIn,H_2}} \approx 40.5. \tag{4.52}$$

This corresponds to the lattice diffusivities

$$\left[D^{LBM}_{H_2},\ D^{LBM}_{TMIn},\ D^{LBM}_{PH_3},\ D^{LBM}_{CH_4}\right] \approx \left[3.49\times10^{-2},\ 2.49\times10^{-2},\ 2.54\times10^{-2},\ 2.62\times10^{-2}\right]. \tag{4.53}$$

Step 4: compute λ, ω^{eff}, and ν_{LBM} for the reactor scale. Using the dominant-H_2 relaxation estimate (Eq. (4.40)) gives

$$\lambda \approx \left(\frac{\phi_{H_2}}{3}\right)\beta\,\chi_{11} \approx 4.65, \qquad \omega^{\mathrm{eff}} = \frac{\lambda}{1+\lambda/2} \approx 1.40, \tag{4.54}$$

and therefore

$$\nu_{LBM} = \frac{1}{3}\left(\frac{1}{\omega^{\mathrm{eff}}} - \frac{1}{2}\right) \approx 7.17\times10^{-2}. \tag{4.55}$$

With the chosen u_{LBM}, the lattice Reynolds number becomes

$$\mathrm{Re}^{LBM} = \frac{u_{LBM}N}{\nu_{LBM}} \approx 7.60,$$

which automatically matches the target physical Reynolds number.

Step 5: micro-layer scale parameters. For the micro-layer simulation, we consider a characteristic shear rate of $\dot{\gamma} = 200\ \mathrm{s}^{-1}$ over a vertical transport distance $L_{\mathrm{sample}} = 500\ \mu\mathrm{m}$. The corresponding characteristic velocity is

$$U_{\mathrm{sample}} = \dot{\gamma} L_{\mathrm{sample}} = 0.10\ \mathrm{m/s}. \tag{4.56}$$

Using the same physical diffusivity as above gives

$$\mathrm{Pe}^{\mathrm{phys}}_{\mathrm{TMIn},\mu} = \frac{U_{\mathrm{sample}} L_{\mathrm{sample}}}{D_{\mathrm{TMIn}}} \approx 1.92 \times 10^{-2}, \qquad \mathrm{Re}^{\mathrm{phys}}_{\mu} = \frac{\rho U_{\mathrm{sample}} L_{\mathrm{sample}}}{\mu} \approx 6.66 \times 10^{-3}. \tag{4.57}$$

Hence the micro-layer problem is in the creeping-flow, diffusion-dominated regime ($\mathrm{Re} \ll 1$ and $\mathrm{Pe} \ll 1$), while the Schmidt number remains unchanged:

$$\mathrm{Sc}^{\mathrm{phys}}_{\mu} = \frac{\mathrm{Pe}^{\mathrm{phys}}_{\mathrm{TMIn},\mu}}{\mathrm{Re}^{\mathrm{phys}}_{\mu}} \approx 2.89.$$

Because $\mathrm{Pe} \ll 1$, we do not use the maximum admissible lattice velocity. Instead we reduce the velocity by three orders of magnitude to allow a finer grid while keeping β in a numerically stable range:

$$u_{\mathrm{LBM},\mu} = \frac{u_{\max}}{1000} \approx 2.03 \times 10^{-6}\ \text{grid/step}, \qquad N_{\mu} = 1000.$$

The required TMIn lattice diffusivity is then

$$D^{\mathrm{LBM}}_{\mathrm{TMIn},\mu} = \frac{u_{\mathrm{LBM},\mu} N_{\mu}}{\mathrm{Pe}^{\mathrm{phys}}_{\mathrm{TMIn},\mu}} \approx 1.06 \times 10^{-1}\ \text{grid/step}. \tag{4.58}$$

The corresponding resistivity multiplier becomes

$$\beta_{\mu} = \frac{1}{D^{\mathrm{LBM}}_{\mathrm{TMIn},\mu} \chi_{\mathrm{TMIn,H_2}}} \approx 9.52. \tag{4.59}$$

This gives the lattice diffusivities

$$\left[D^{\mathrm{LBM}}_{\mathrm{H_2},\mu},\ D^{\mathrm{LBM}}_{\mathrm{TMIn},\mu},\ D^{\mathrm{LBM}}_{\mathrm{PH_3},\mu},\ D^{\mathrm{LBM}}_{\mathrm{CH_4},\mu}\right] \approx \left[1.49 \times 10^{-1},\ 1.06 \times 10^{-1},\ 1.08 \times 10^{-1},\ 1.11 \times 10^{-1}\right]. \tag{4.60}$$

The corresponding relaxation quantities are

$$\lambda_{\mu} \approx \left(\frac{\phi_{\mathrm{H_2}}}{3}\right) \beta_{\mu} \chi_{11} \approx 1.09, \qquad \omega^{\mathrm{eff}}_{\mu} = \frac{\lambda_{\mu}}{1 + \lambda_{\mu}/2} \approx 0.707, \tag{4.61}$$

and therefore

$$\nu_{\mathrm{LBM},\mu} = \frac{1}{3}\left(\frac{1}{\omega^{\mathrm{eff}}_{\mu}} - \frac{1}{2}\right) \approx 3.05 \times 10^{-1}. \tag{4.62}$$

The resulting lattice Reynolds number is

$$\mathrm{Re}^{\mathrm{LBM}}_{\mu} = \frac{u_{\mathrm{LBM},\mu} N_{\mu}}{\nu_{\mathrm{LBM},\mu}} \approx 6.66 \times 10^{-3},$$

again matching the target physical Reynolds number.

In summary, the two simulations use the same composition-dependent transport scaling ϕ, while different choices of $(N, u_{\mathrm{LBM}}, \beta)$ realize the reactor-scale advection–diffusion regime and the micro-layer diffusion-dominated regime within the same MS–LBM framework.

4.3. Reactor-scale simulation

We next simulate the reactor-scale flow and concentration fields in a simplified 2D cross-section. The species set is $\{\mathrm{H_2}, \mathrm{TMIn}, \mathrm{PH_3}, \mathrm{CH_4}\}$. A 38 mm wafer is placed at the bottom boundary and acts as a reactive surface that consumes TMIn and PH_3 and produces CH_4 with a fixed stoichiometric ratio (implemented through coupled Robin/source boundary conditions).

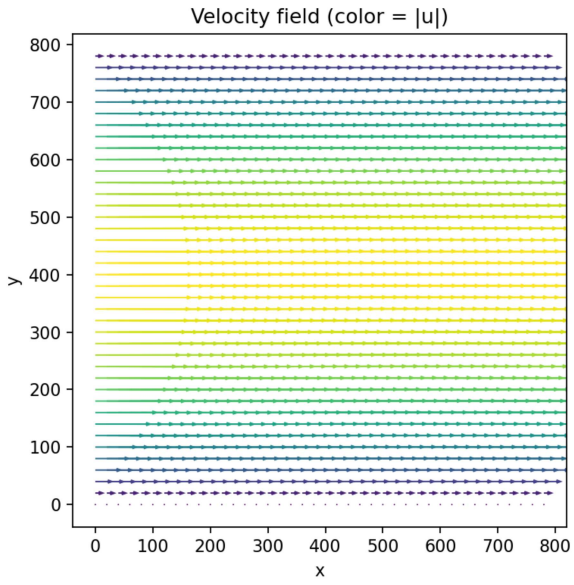


Figure 4.1: Velocity vectors in the 2D reactor cross-section after 12×10^6 steps. The maximum velocity at the center is 2×10^{-3}grid/step The flow remains laminar and follows the prescribed Couette–Poiseuille inflow profile (Eq. (4.63)), with no-slip at the top and bottom boundaries. Horizontal and vertical axes are x and y positions respectively, in the unit of lattice (1 lattice = 500μm).

4.3.1. Geometry and boundary conditions

The computational domain is the part of the reactor where the 38 mm quarter-sector wafer is located. It is discretized in to a 800×800 grid, with the wafer located at the bottom at grid $200 \to 600$. At the left inlet, the mixture composition is prescribed and a fully developed laminar inflow is imposed; in general this can be expressed in Couette–Poiseuille form, and in the present setup we use the parabolic profile

$$u_x(y) = 4u\frac{y}{H}\left(1 - \frac{y}{H}\right), \tag{4.63}$$

where u sets the mean speed. At the right outlet we apply a zero-normal-gradient outflow for populations entering from the boundary. The top boundary is a stationary no-slip wall. The bottom boundary is segmented: the reactive wafer section uses a Robin condition to model uptake, while the remaining bottom wall is non-reactive and treated as reflecting/no-slip.

4.3.2. Flow field

Figure 4.1 shows the velocity field after 12×10^6 time steps. The computed streamwise velocity profile is consistent with the imposed inflow (Eq. (4.63)): the no-slip condition enforces $u_x = 0$ at the top and bottom boundaries, while the core flow retains the expected rightward through-flow.

Importantly, we do not observe spurious recirculation or “backflow” layers near the walls at late times. This indicates that (i) the inlet profile is compatible with the wall boundary conditions, and (ii) the simple zero-gradient outlet does not introduce noticeable reflections into the domain for the present low-Re regime. Within numerical tolerance, the bulk flow is steady, and we do not observe any significant diffusion-induced advection (i.e. species redistribution does not measurably perturb the mixture momentum under these dilute-precursor conditions).

4.3.3. Concentration fields

Figure 4.2 summarizes the steady-state concentration fields for the four-species system Because H_2 is the dominant carrier gas, its concentration remains essentially uniform throughout the domain. PH_3 is supplied in large excess relative to TMIn (V/III ≈ 435), so although PH_3 is consumed at the wafer, its *relative* depletion in the bulk is small compared with TMIn; correspondingly, the strongest precursor

gradient is observed for TMIn.

Over the reactive wafer segment, TMIn and PH_3 are removed through the Robin-type surface boundary condition, creating a near-surface depletion layer. In contrast, CH_4 is produced at the same segment and therefore forms a near-wall enrichment layer. Downstream of the wafer, the CH_4 field is advected to the right and simultaneously diffuses upward, producing a ”fume” that marks the reaction zone and its wake.

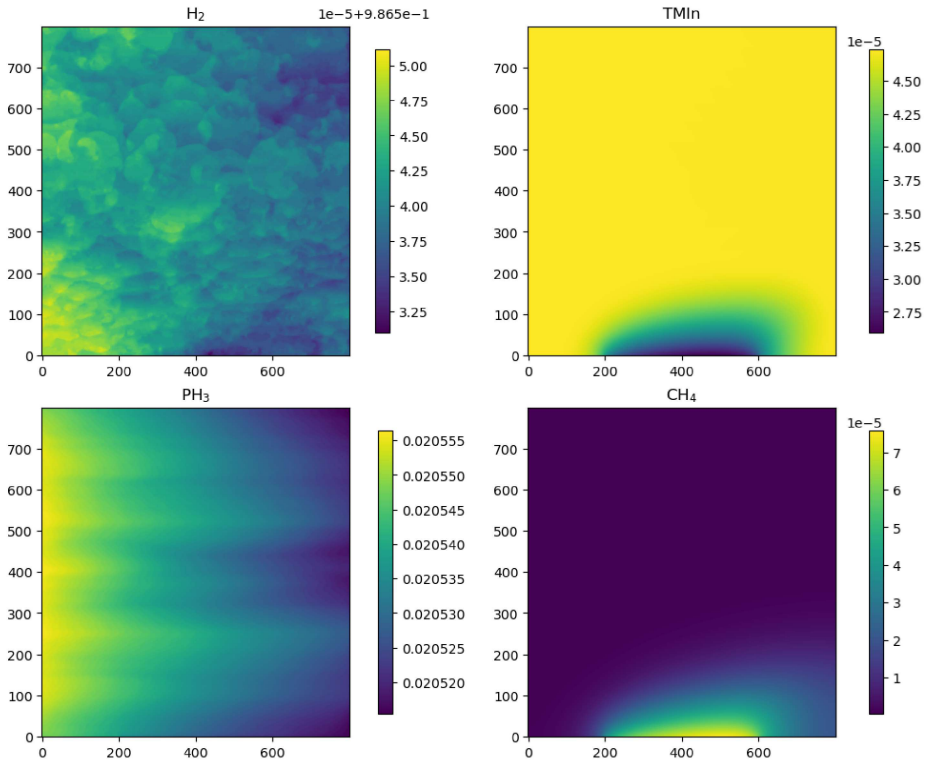
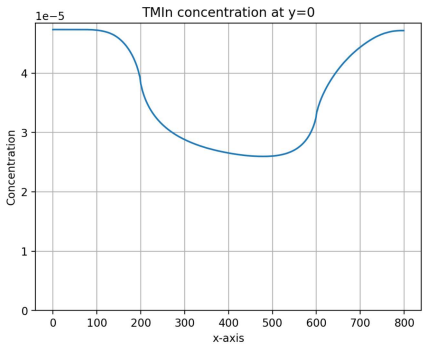
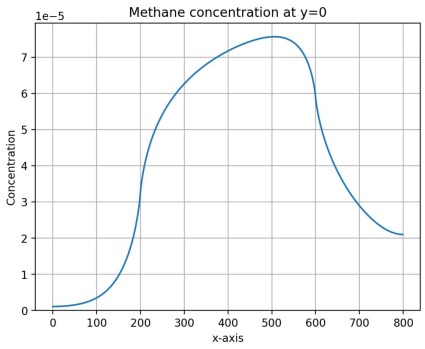


Figure 4.2: Steady-state concentration fields of the four species after 12×10^6 steps. TMIn shows the strongest depletion above the wafer due to surface consumption, while CH_4 is generated at the wafer and transported downstream by advection and diffusion. Horizontal and vertical axes are x and y positions respectively, in the unit of lattice (1 lattice = 500μm).

To highlight the streamwise bias close to the surface, Figure 4.3 shows the near-surface concentrations sampled at the first fluid node above the bottom boundary along the wafer direction. TMIn is highest on the upstream (left) side of the wafer and decreases along the flow direction as the boundary layer develops and precursor is progressively consumed. CH_4 exhibits the opposite trend: it accumulates downstream because it is continuously generated at the wafer and advected to the right. Although the absolute CH_4 fraction is small (and thus has negligible feedback on the overall mixture properties and flow), its spatial distribution provides a clear qualitative indicator that the implemented reactive boundary condition produces the expected source–sink coupling between reactants (TMIn/PH_3) and product (CH_4).



(a) TMIn



(b) CH_4

Figure 4.3: Near-surface concentrations sampled at the first fluid node above the wafer, plotted along the wafer direction. TMIn decreases downstream due to progressive depletion by surface consumption, while CH_4 increases downstream due to continuous production and advection.

4.4. Micro-layer scale simulation

The reactor-scale simulation (Section 4.3) provides near-wafer gas-phase compositions on a coarse streamwise grid (0.5 mm spacing). We use these near-surface concentrations as the inflow composition for a micro-layer simulation resolving transport above a patterned mask on the wafer. The objective is to quantify how micro-scale masking perturbs the local reactant supply and the resulting growth-rate (thickness) pattern.

4.4.1. Geometry and boundary conditions

The micro-layer domain is a $400\ \mu$m long by $200\ \mu$m tall 2D cross-section above the wafer surface. A patterned mask occupies the central portion of the bottom boundary, leaving open (unmasked) windows where growth can proceed. The mask layout is shown in Fig. 4.4.

Flow enters from the left boundary with a prescribed linear (Couette-type) streamwise velocity profile. The top wall translates rightward at a constant speed, enforcing shear consistent with the inlet profile. The right boundary is treated as a convective/zero-gradient outflow. On the bottom boundary, reactive Robin boundary conditions are applied only on the unmasked windows (openings), while masked regions are treated as inert wall (no reaction / no-flux for species). The reactive windows consume TMIn and PH_3 and produce CH_4 with the same fixed stoichiometric coupling used at reactor scale.

4.4.2. Diffusion-dominated transport at micro-scale

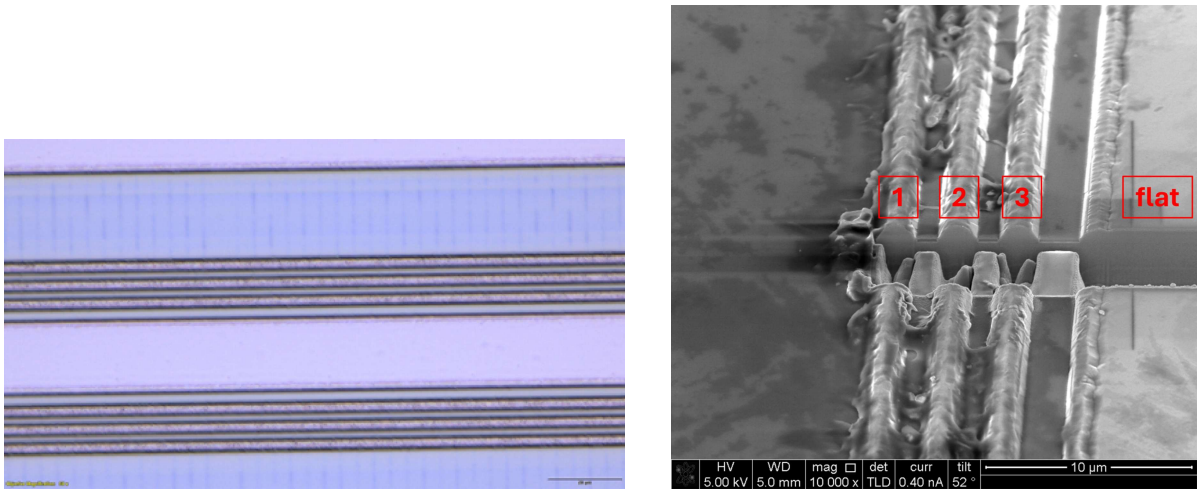
For the present operating point, transport in the micro-layer is strongly diffusion-dominated, with $\mathrm{Pe} \sim 2 \times 10^{-2}$. A simple magnitude estimate illustrates why advection is negligible. Using characteristic lattice-unit values from the simulation, $U \sim 10^{-6}$ (grid/step), $D \sim 0.1$ (grid^2/step), and a representative length scale $L \sim 10^3$ (grid), the ratio of advective to diffusive transport scales as

$$\frac{|u \cdot \nabla c|}{|D \nabla^2 c|} \sim \frac{UL}{D} = \mathrm{Pe} \sim 10^{-2}, \tag{4.64}$$

i.e. advection contributes only at the percent level relative to diffusion.

This behavior is also consistent with the dilute-precursor regime: TMIn is the limiting component and is an extremely small fraction of the mixture (PH_3 is only a few percent of the carrier, and TMIn is smaller still by the V/III ratio). Consequently, concentration redistribution of the precursors does not drive any appreciable mixture motion (no measurable diffusion-induced advection).

Figure 4.5 shows the steady flow field. Although the top wall motion imposes shear, velocities near the wafer are extremely small. Figure 4.6 shows the corresponding steady concentration fields. In contrast to the reactor-scale results (Fig. 4.2), there is no pronounced downstream ”product fume” in



(a) (b)

Figure 4.4: Micro-mask configuration used in the micro-layer simulations. (a) Mask opening design: two primary stripes (each ≈ 40 μm wide) separated by a ≈ 20 μm gap, plus three narrow slots near the inner mask edge (each ≈ 2–3 μm wide, with comparable spacing). (b) InGaAsP growth characterized by focused ion beam electron microscopy. The material grown at the center gap is In richer compared to that grown on the slots on the mask.

the micro-layer domain, further indicating that diffusion controls species redistribution at this scale.

4.4.3. Comparison with a diffusion-only model

Given $\mathrm{Pe} \ll 1$, it is natural to compare the LBM results against a diffusion-only model (VPD) obtained by solving $\nabla^2 c = 0$ (steady Laplace problem) with the same top and bottom boundary conditions as the LBM case. Figure 4.7 compares the resulting normalized concentration at the bottom (here used as a proxy for the local growth-rate pattern driven by reactant supply into the mask openings). The LBM reproduces the qualitative trend predicted by the diffusion model, but yields overall higher normalized thickness and a weaker contrast between the narrow edge slots. Despite the rightward flow, the concentration profile is almost symmetric, with only a negligible amount higher on the left side compare to the right. Compared to figure 4.3a where the concentration is biased by the flow direction, this further demonstrated at micro-layer scale the transport is dominated by diffusion.

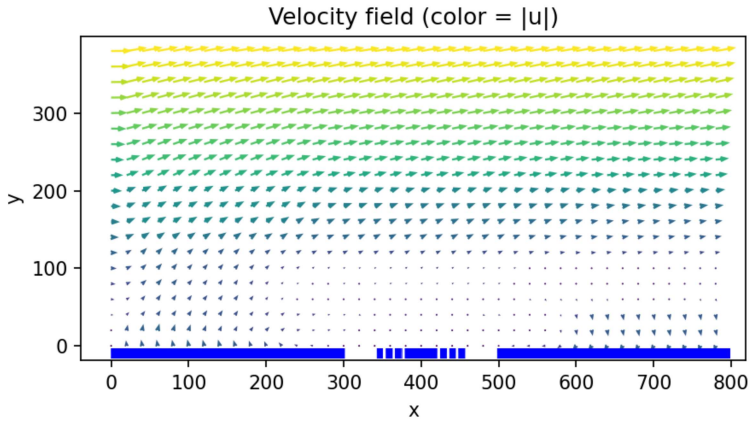


Figure 4.5: Velocity vectors in the micro-layer domain. The top wall moves rightward (speed $\approx 3.3 \times 10^{-6}$ in lattice units), generating a weak Couette shear. The near-wafer velocity is negligible, consistent with $\mathrm{Pe} \ll 1$. The blue segment marks the unmasked (reactive) window(s) on the bottom boundary. Horizontal and vertical axes are x and y positions respectively, in the unit of lattice (1 lattice = 0.5μm).

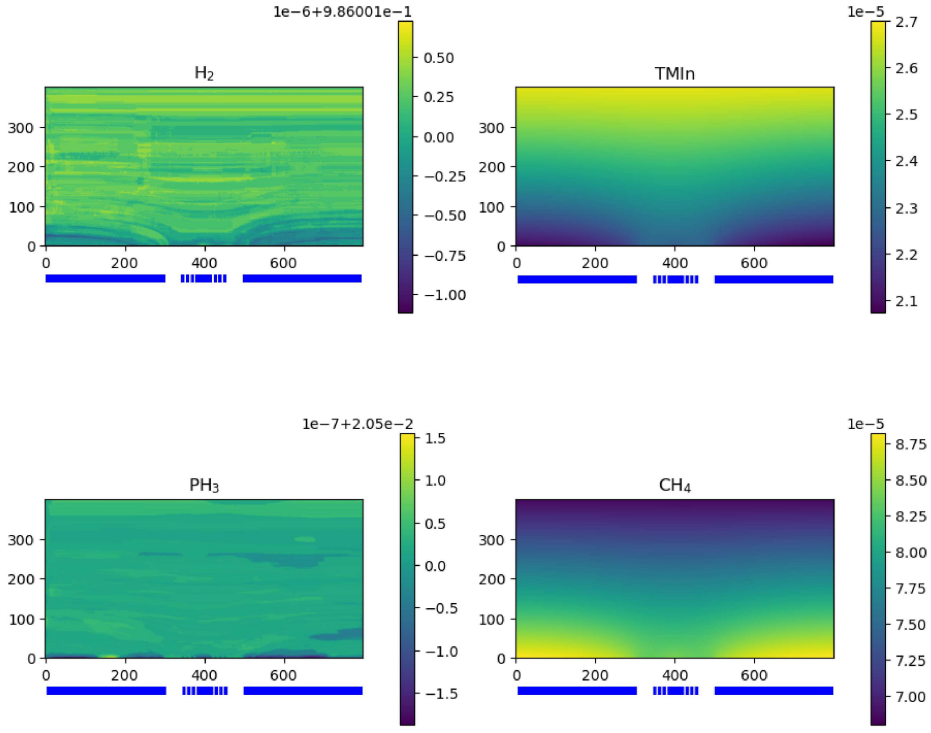


Figure 4.6: Steady concentration fields in the micro-layer simulation for the four-species system $\{\mathrm{H_2}, \mathrm{TMIn}, \mathrm{PH_3}, \mathrm{CH_4}\}$. The absence of a strong downstream product "fume" (as seen at reactor scale) indicates diffusion-dominated transport at micro-scale. The blue segment marks the unmasked (reactive) region on the bottom boundary. Horizontal and vertical axes are x and y positions respectively, in the unit of lattice (1 lattice = 0.5μm).

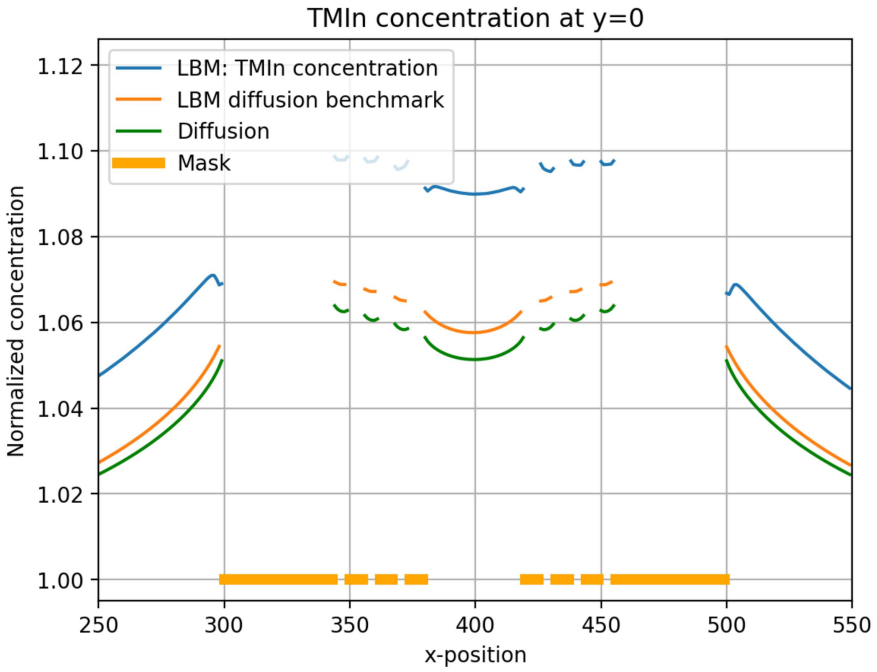


Figure 4.7: Normalized concentration (growth-rate proxy) near the mask openings for the same geometry and boundary conditions, comparing three models: four-species LBM (InP case), two-component tracer-diffusion LBM benchmark, and a diffusion-only reference solution (steady Laplace). The LBM diffusion benchmark showed result very close to the diffusion-only curve, confirming correct diffusion-dominated transport for an appropriate parameter regime, whereas the four-species LBM follows the same qualitative spatial trend but predicts systematically higher normalized thickness and weaker contrast near the narrow edge slots. The x axis is in the unit of lattice (1 lattice = 0.5μm).

To determine whether the discrepancy originates from diffusion-driven advection or from accuracy of the LBM implementation, we performed an additional two-component tracer-diffusion run (background + tracer). In this test, we used $M_1 = 1$, $M_2 = 1.5$, $\phi_1 = 1$, and $\beta = 5$ (parameter set that better matches the theoretical diffusion rate in the diffusion benchmark of Section 6.1). We also increased the tracer fraction (e.g. 99.9% and 0.1% tracer) to amplify any potential flow coupling and thereby stress-test the transport model. As shown in Fig. 4.7, the LBM matches the diffusion-only solution very closely in this controlled setting. This indicates that the implementation is capable of accurate diffusion prediction, while the reduced accuracy in the four-species InP parameter set is more likely attributable to the particular parameter regime. While the diffusion induced advection (supplementary figure A.3)) is more significant compared to InP case in figure 4.5, it is still at the order of magnitude of 10^{-6}grid/step, which do not affect the boundary concentration.

In contrast, the four-species InP case reproduces the qualitative spatial trend predicted by the diffusion-only solution, but predicts systematically higher normalized thickness. At the same time, it exhibits a weaker contrast near the narrow edge slots compared with the diffusion-only (and tracer-benchmark) result. This combination suggests that the remaining discrepancy is more likely tied to the specific multi-component parameter regime used for the InP case than to unmodeled advection in the micro-layer domain. We suspect this is inaccuracy happens when the ratio between molecular mass is large, or ϕ is away from 1, but we did not have time to fully quantify this inaccuracy. We discussed more on the diffusion accuracy in section 6.1.

4.4.4. Comparison with experimental result

Here we compare our InP growth simulation with experimental result of InGaAsP growth. Though growth of InGaAsP is limited by both TMIn and TMGa (Table A2 in Supplementary information II), under a diffusion-dominated regime the relative indium-supply pattern is still expected to be qualitatively comparable between the two cases.

Figure 4.4b shows the InGaAsP sample growth characterized by electron microscope. The material

growth on the slots (marked with position 1, 2, and 3) on the mask appears irregular. A plausible explanation is locally unstable or non-optimal reactant composition during crystallization, which can cause defective lattice formation. Consistent with this interpretation, both our simulation and the VPD model predict strong local variations in indium supply at the narrow slots (figure 4.7), which could contribute to the observed irregular growth.

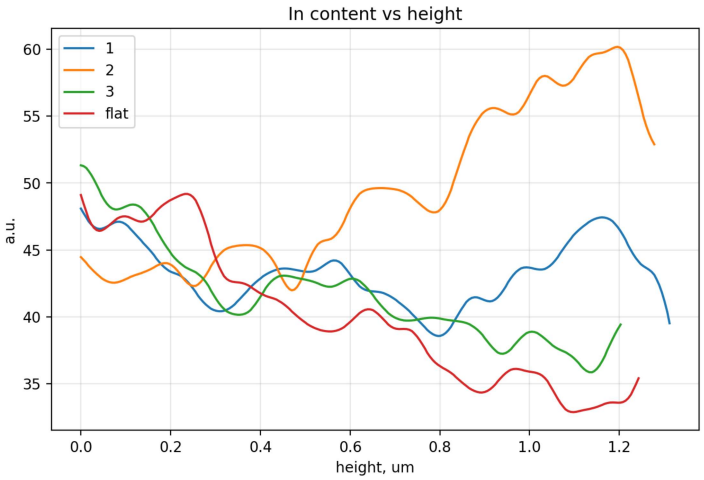


Figure 4.8: EDX measurement of In composition vs height on four locations of the sample, marked on figure 4.4b. The x-axis is the scan position, with $x = 0\mu$m at the bottom of the sample and $x \approx 1.2\mu$m at the top. The raw data is smoothed with Gaussian filter.

Energy-Dispersive X-ray Spectroscopy (EDX) further characterized the indium composition of InGaAsP growth. Figure 4.8 shows the In content versus height at four locations (marked on figure 4.4b). From $x = 0\mu$m (bottom) to $x \approx 0.6\mu$m, the In content is relatively similar across all four locations. From $x > 0.6\mu$m, the flat region (the wide center gap between two main masks) shows the lowest indium content, followed by position 3 and 1, which is consistent with our model trend. However, position 2 shows a substantially higher value, contradicting the model prediction.

At present, we cannot explain this discrepancy within the scope of the Maxwell–Stefan diffusion–advection transport model used here.The deviation may arise from effects not included in the model. Model limitations and future development are discussed in Chapter 5.

5

Discussion

5.1. Limitations of the current implementation

Combining the reactor-scale result that the near-surface concentration depends on the flow direction (Fig. 4.3) with the micro-layer simulations where transport is diffusion-dominated (figure 4.7), we infer a multi-scale transport mechanism: advection delivers reactants into the millimeter-scale region above the wafer, while the final delivery to the surface occurs primarily by micrometer-scale diffusion.

Despite including a coupled multi-species flow description, the present model still relies on major simplifying assumptions, which likely limit its quantitative accuracy. The most significant simplification is the *isothermal, constant-property* treatment. In the reactor, the carrier gas enters near room temperature, whereas the susceptor/wafer is heated to approximately $650\,^{\circ}\mathrm{C}$. Such a large temperature gradient can strongly modify viscosity and diffusion coefficients, and it can introduce density variations that drive buoyancy-induced convection [19]. These coupled effects can alter both the near-surface concentration field and the effective mass-transfer rate, and may contribute to the mismatch in the near-surface absorption slope discussed in Section 5.2.

A second key limitation is the absence of detailed *gas-phase and surface chemistry*. In the current model, TMIn and PH_3 are converted into InP and CH_4 at the surface via a single-step reaction, whereas in practice the chemistry proceeds through multiple gas-phase decomposition reactions and surface-mediated steps, with strong sensitivity to the local temperature and composition [20, 21]. Neglecting these pathways effectively folds reaction kinetics into an oversimplified boundary sink, which can bias predicted depletion profiles and growth-rate proxies.

To extend the present framework toward a quantitatively predictive reactor model, we propose the following development roadmap: introduce thermal coupling by solving for the temperature field and using T-dependent viscosity and binary diffusion coefficients; enable variable-density effects and buoyancy forcing to capture thermally driven convection in the near-wafer region; and replace the single-step surface sink with a temperature-dependent, multi-step chemistry model that includes both gas-phase decomposition and surface reaction pathways. Together, these upgrades will allow the model to resolve the coupled interplay of flow, heat transfer, transport, and chemistry that governs selective-area growth.

The complexity of this extended simulation can be much more difficult compared to the current implementation. In 2021, Sawant et al.[22] developed the first LBM framework that handles chemistry, thermal effects, compressibility, and Maxwell-Stefan diffusion. They used Cantera [23], a open source chemical kinetics solver, to get parameters needed to couple LBM, including reaction kinetics, viscosity, thermal conductivity, and the Maxwell-Stefan diffusivity. Despite no code publicly available, recreating the algorithm and simulating validation cases based in their paper is doable. Despite the difference in implementation, the codebase we developed in this project still serves a good starting point for understanding the species-diffusion part.

5.2. Discrepancy on absorption rate

The biggest discrepancy between experimental data, our simulation results, and prior literature (VPD model fit) is the surface uptake (absorption) rate. To reproduce the magnitude of thickness variations of SAG [1, 3, 24], the model require much stronger effective absorption rate than the estimation based on measured InP growth rate.

In section 4.1.5 equation (4.31), we estimated from InP growth rate (180 $\AA$/min) that the TMIn absorption flux to be $J_{\text{TMIn}} \approx 9.90 \times 10^{-6}\ \text{mol}/(\text{m}^2\,\text{s})$. According to Fick's first law

$$J = -D\frac{\partial c}{\partial x} \tag{5.1}$$

the concentration gradient of TMIn at the reaction boundary is

$$\frac{\partial c_{\text{TMIn}}}{\partial x} = \frac{J_{\text{TMIn}}}{D_{\text{TMIn, H}_2}} = 3.8 \times 10^{-3}\ (\text{mol/m}^3)/\text{m} \tag{5.2}$$

Both equation (5.2) and (3.46) can be normalized by the inlet concentration $c_{\infty,\text{TMIn}} = 6.0\times10^{-5}\text{mol/m}^3$:

$$c_{norm} = \frac{c_{\text{TMIn}}}{c_{\infty,\text{TMIn}}} \tag{5.3}$$

This gives a normalized boundary gradient

$$\frac{\partial c_{\text{norm}}}{\partial x} = 65/\text{m} \tag{5.4}$$

We set the parameters of the Robin condition in reactor-scale simulation to match the experimental reaction rate. According to the Robin boundary condition (equation 2.4 and 3.46), and rate of reaction (equation 4.32),

$$\frac{b_2}{b_1} = \frac{D_{\text{TMIn}}}{k_{\text{rxn}}} = 1.6 \times 10^{-2}\ \text{m} = 168\,\text{grid} \tag{5.5}$$

(In practice the boundary condition (section 3.5.3) implementation introduces numerical error: we used $\frac{b_2}{b_1}$ to 130 grid, and measured from simulation result $\frac{b_2}{b_1} = 230$ grid, with $\frac{\partial c_{\text{norm, measured}}}{\partial x} = 49/\text{m}$.)

However, **this value contradicts against VPD model** suggested in the literature, which fitted "effective diffusion length" $\frac{D}{k}$ to measured thickness profiles of growth [1, 3, 24]. Their data fitting showed the $\frac{D}{k}$ is around 15μm to 85μm, which is 10^3 **times higher than the value estimated from the growth rate**.

We speculate this huge difference in $\frac{D}{k}$ parameter is due to a wrong assumption in our model: TMIn does not directly convert to InP in a single-step reaction at the surface. Instead, the **transport is limited by a low-concentration intermediate product in the multi-step gas phase chemical reaction**. The intermediate product that directly leads the growth of InP is generated in high-temperature region near the hot wafer, at low concentration. For a fixed growth flux $J = kc$, a smaller concentration c implies a large average drift velocity k, which in turn produces a much smaller $\frac{D}{k}$, consistent with VPD fitting.

This interpretation also highlights a limitation of the VPD approach itself: while the VPD model predicts the growth pattern and effective diffusion length by fitting experimental data, it overlooks the actual transport and reaction process. This also confirms the necessity of a thermal and gas-phase reaction coupled flow simulation. Only when the model simulates the temperature field of the system, the temperature-dependent gas-phase chemical reaction can be modeled, generating intermediate product that directly affects the local growth rate. In that setting, reactor flow is expected to matter more than in the current transport-only model, because it will influence not only reactant delivery but also the temperature field that controls the chemistry producing the actual growth-limiting species.

6

Supplementary information I: LBM validation

6.1. Tracer-diffusion validation

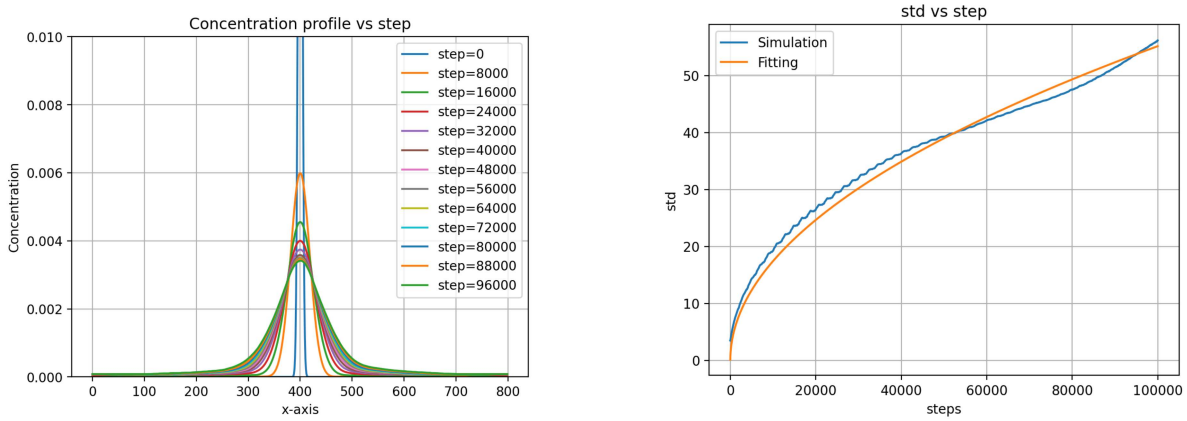
To validate the diffusive transport implemented in our LBM framework, we simulated the diffusion of a dilute TMIn tracer in an H_2 background. The TMIn concentration is initialized as a narrow Gaussian peak centered along the x-axis (Fig. 6.1a), while the velocity field is set to zero everywhere ($\mathbf{u} = \mathbf{0}$). We use $\beta = 50$ and $\phi_{H_2} = 0.5$, which corresponds to a target lattice diffusivity $D = 0.0201\ \mathrm{grid}^2/\mathrm{step}$, consistent with the parameter regime used in the reactor-scale simulations.

As the simulation evolves, the concentration profile remains approximately Gaussian and broadens in time. We quantify this spreading by fitting the standard deviation $\sigma(t)$ of the TMIn profile and comparing it to the analytical solution of the 1D diffusion equation for an initially Gaussian distribution,

$$\sigma^2(t) = \sigma_0^2 + 2Dt, \tag{6.1}$$

which reduces to $\sigma(t) \approx \sqrt{2Dt}$ when the initial width σ_0 is sufficiently small. Accordingly, we fit σ^2 as a linear function of time (Fig. 6.1b) and extract the effective diffusivity from the slope.

During the first 2×10^4 time steps, the broadening follows the expected $\sigma^2 \propto t$ scaling, yielding $D \approx 0.019\ \mathrm{grid}^2/\mathrm{step}$, in good agreement with the prescribed value. At later times, the apparent spreading slows (Fig. 6.1b); by $t = 5 \times 10^4$ steps, the fitted value decreases to $D \approx 0.01529\ \mathrm{grid}^2/\mathrm{step}$.



(a) (b)

Figure 6.1: Tracer-diffusion validation: (a) initial Gaussian tracer profile and (b) extracted standard deviation over time steps.

This late-time deviation indicates a departure from the ideal infinite-domain diffusion model, and may reflect finite-domain/boundary effects and/or discretization-related artifacts.

6.2. Kinematic viscosity verification from decaying Fourier modes

To verify that the lattice Boltzmann implementation reproduces the intended kinematic viscosity, we consider two canonical unsteady benchmarks with closed-form exponential decay [25]: (i) a transverse shear-wave decay and (ii) the two-dimensional Taylor–Green vortex (TGV). Both are single-(or few-)mode periodic flows for which incompressible theory predicts that the spatial mode shape is preserved and the amplitude decays exponentially with a rate set by ν and the wavenumber k.

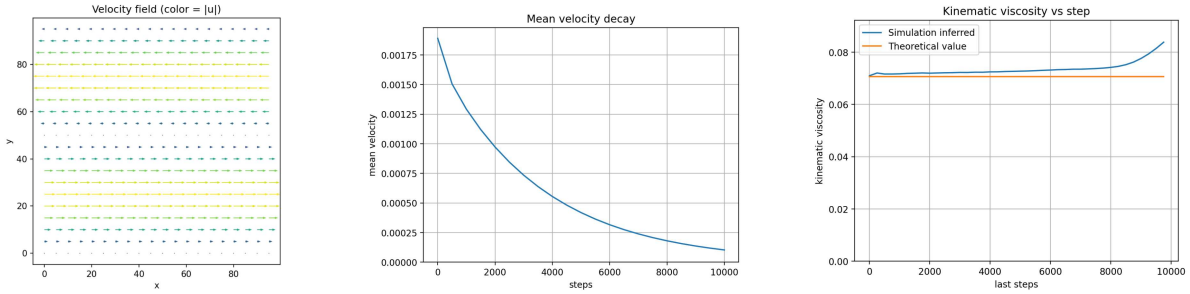
6.2.1. Shear-wave decay

We simulate a single-component isothermal fluid on a periodic domain of size $L_x \times L_y$ discretized on an $N_x \times N_y$ lattice (lattice units $\Delta x = \Delta t = 1$). The initial density is uniform, $\rho(\mathbf{x}, 0) = \rho_0$, and the initial velocity is a single transverse mode as shown in figure 6.2a,

$$u_x(\mathbf{x}, 0) = 0, \qquad u_y(\mathbf{x}, 0) = u_0 \sin(k\,\xi), \qquad k = \frac{2\pi}{L_\xi}, \tag{6.2}$$

where ξ denotes the coordinate along which the sinusoidal perturbation is imposed. In our implementation we use $\xi = y$ so that $L_\xi = L_y = N_y$. The incompressible prediction is

$$u_y(\mathbf{x}, t) = u_0 \sin(k\,\xi)\, \exp\left(-\nu k^2 t\right). \tag{6.3}$$



(a) (b) (c)

Figure 6.2: Shear-wave benchmark: (a) shear-wave velocity field, (b) measured mode-amplitude proxy $A(t)$ showing exponential decay consistent with Eq. (6.3), and (c) inferred viscosity from the from the steps till the end of the simulation vs theoretical value.

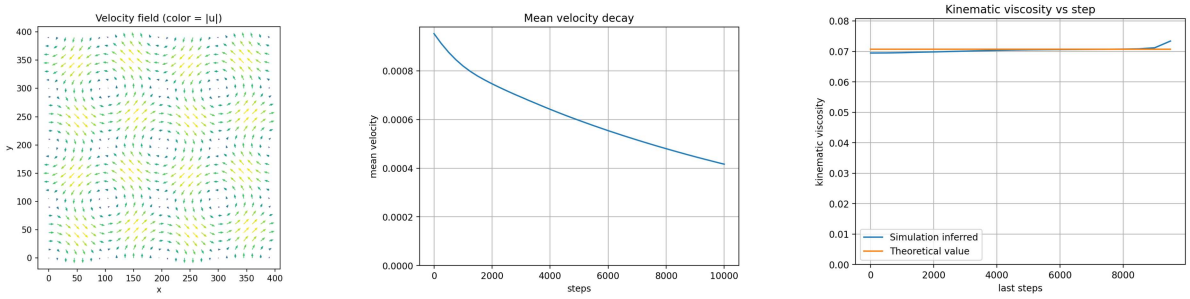
6.2.2. Taylor–Green vortex decay

As a second benchmark we simulate the 2D Taylor–Green vortex on a periodic square domain $[0, L]^2$ with $k = 2\pi/L$. The initial flow field, as shown in figure 6.3a is

$$u_x(x, y, 0) = U_0 \cos(kx) \sin(ky), \qquad u_y(x, y, 0) = -U_0 \sin(kx) \cos(ky). \tag{6.4}$$

In incompressible theory the spatial structure is preserved and the amplitude decays as

$$u_x(x, y, t) = U_0 \cos(kx) \sin(ky)\, \exp\left(-2\nu k^2 t\right), \qquad u_y(x, y, t) = -U_0 \sin(kx) \cos(ky)\, \exp\left(-2\nu k^2 t\right). \tag{6.5}$$



(a) (b) (c)

Figure 6.3: Taylor–Green vortex benchmark: (a) flow field snapshot, (b) measured mode-amplitude proxy $A(t)$ showing exponential decay consistent with Eq. (6.3), and (c) inferred viscosity from the from the steps till the end of the simulation vs theoretical value.

6.2.3. Viscosity measurement and estimation

For both benchmarks, the analytical solutions (6.3) and (6.5) imply that a suitable scalar amplitude $A(t)$ decays exponentially:

$$A(t) = A(0)\,\exp\!\left(-m\nu k^2 t\right), \qquad m = \begin{cases} 1, & \text{shear wave,} \\ 2, & \text{Taylor–Green vortex.} \end{cases} \tag{6.6}$$

In principle, $A(t)$ can be obtained by projecting the velocity field onto the corresponding Fourier basis function. In our implementation we use a simpler proxy that is proportional to the mode amplitude for these single-mode initializations:

- Shear wave: $A(t_n) = \text{mean}_{x,y}(|u_y(x,y,t_n)|)$.
- TGV: $A(t_n) = \text{mean}_{x,y}(\sqrt{u_x(x,y,t_n)^2 + u_y(x,y,t_n)^2})$.

These definitions yield $A(t) \propto \exp(-m\nu k^2 t)$ as long as the flow remains dominated by the initialized mode. The decay profiles are shown in figures 6.2b and 6.3b.

Let $A_0 = A(t_0)$ be the initial amplitude and $A_i = A(t_i)$ the amplitude at the last recorded frame. If frames are stored every Δt_{out} time steps, then $t_i = i\,\Delta t_{\text{out}}$. From (6.6) we obtain the two-point estimator

$$\nu_{\text{num}} = -\frac{1}{mk^2 t_i}\,\ln\!\left(\frac{A_i}{A_0}\right), \qquad t_i = i\,\Delta t_{\text{out}}. \tag{6.7}$$

with $m = 1$ for the shear-wave case and $m = 2$ for the TGV.

In figures 6.2c and 6.3c, the inferred ν_{num} from the profile of last x steps in the simulation, from (6.7) is compared with the viscosity implied by the LBM relaxation parameters, ν_{LBM} (Eq. (4.38).

We noticed if the ν_{num} inference is from the steps exclude the very first $1000 \sim 2000$ steps, its value is very close to the theoretical value. However, during the initial steps, where f_i is initialized from equilibrium states of designated velocities, the behavior of the system deviates from the prediction.

7

Generative AI usage

During this project, AI coding tools were used as programming aids for certain auxiliary tasks, including optimizing the ”solve_ms_fluxes” function, improving plots, and fixing package compatibility. It has also been used for improving the clarity and language of the project report. The primary scientific implementations of the LBM algorithm and result analysis were authored without AI-generated code.

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Supplementary information II

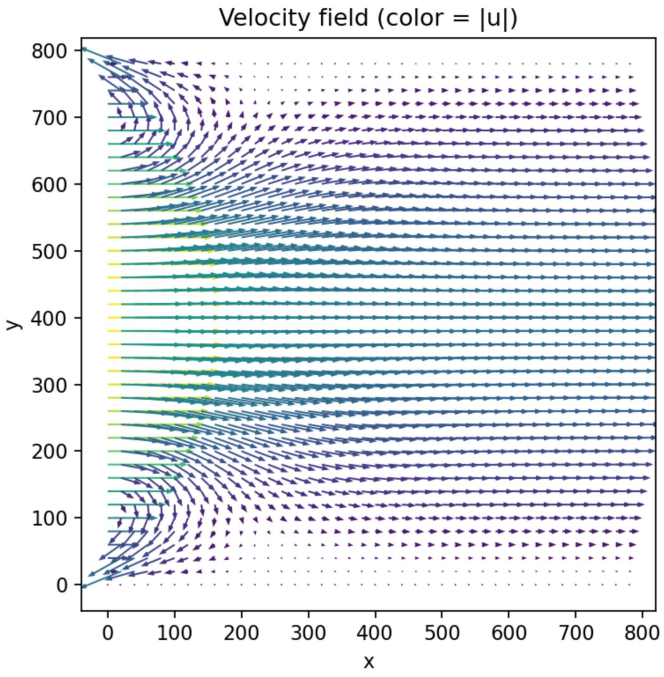


Figure A.1: With the enforced boundary to equilibrium inlet, the flow goes backwards near the inlet, due to the inlet can not provide enough ”pressure”.

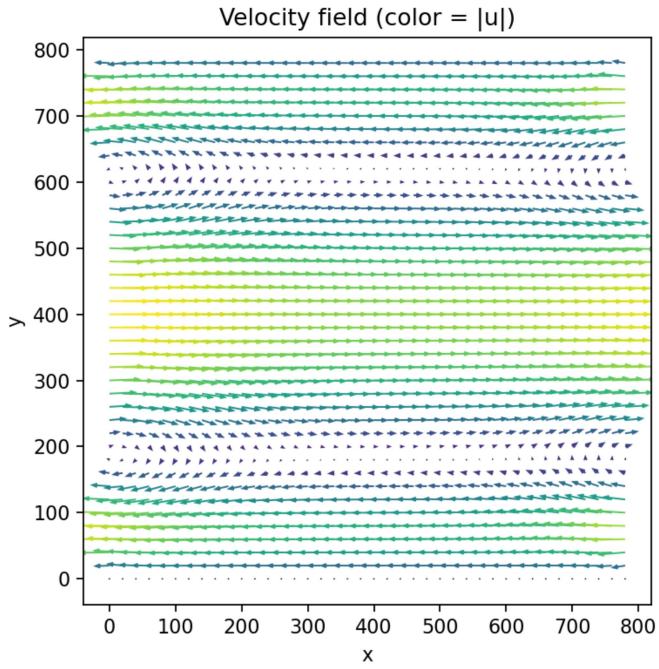


Figure A.2: With NEE inlet boundary condition, flow exit the inlet backwards.

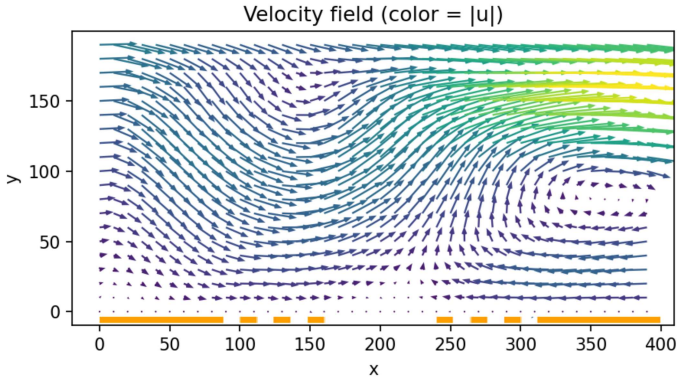


Figure A.3: Flow field of the LBM diffusion benchmark.The diffusion of the tracer induce flow.

Table A.1: General reactor conditions and growth rates for the reference InP and InGaAsP runs.

Parameter	InP	InGaAsP (Run #4845)
Reactor volume above susceptor	1.23 L (1.23×10^{-3} m^3)	
Reactor pressure	100 mbar	100 mbar
Reactor temperature	650 °C	650 °C
Growth rate	180 Å min^{-1}	222 Å min^{-1}

Table A.2: Gas flows, estimated partial pressures, and flow-derived precursor ratios for the InP and InGaAsP runs.

Quantity	InP	InGaAsP (Run #4845)
Carrier gas		
H_2 flow	15 000 sccm	15 000 sccm
H_2 partial pressure	97.953 mbar	97.903 mbar
Group-V precursors		
PH_3 flow	306.3 sccm	306.3 sccm
PH_3 partial pressure	2.042 mbar	2.0417 mbar
AsH_3 flow	–	8.32 sccm
AsH_3 partial pressure	–	0.0555 mbar
Group-III precursors		
TMIn pure flow	0.704 sccm	0.704 sccm
TMIn partial pressure	4.693×10^{-3} mbar	4.693×10^{-3} mbar
TMGa pure flow	–	0.274 sccm
TMGa partial pressure	–	1.828×10^{-3} mbar
Derived quantities		
Gas-phase V/III ratio	435.1	321.7
Precursor-flow ratio In:Ga:As:P[a]	$1 : 0 : 0 : 435.1$	$1 : 0.389 : 11.82 : 435.1$

Partial pressures are estimated from the inlet flow ratios using the convention provided in the process data.

For the InP case, the corrected TMIn partial pressure is 4.693×10^{-3} mbar, consistent with a TMIn pure flow of 0.704 sccm at 100 mbar. The value 0.0469 appearing in the raw notes is inconsistent with the stated TMIn flow and was therefore not used.

[a] Flow-normalized precursor ratio, normalized to TMIn = 1. This is a precursor supply ratio, not the final solid composition.